جامعة الملك سعود
عمادة شؤون الطلاب
صندوق الطلاب

King Saud University
College of Engineering
Electrical Engineering Departmemt

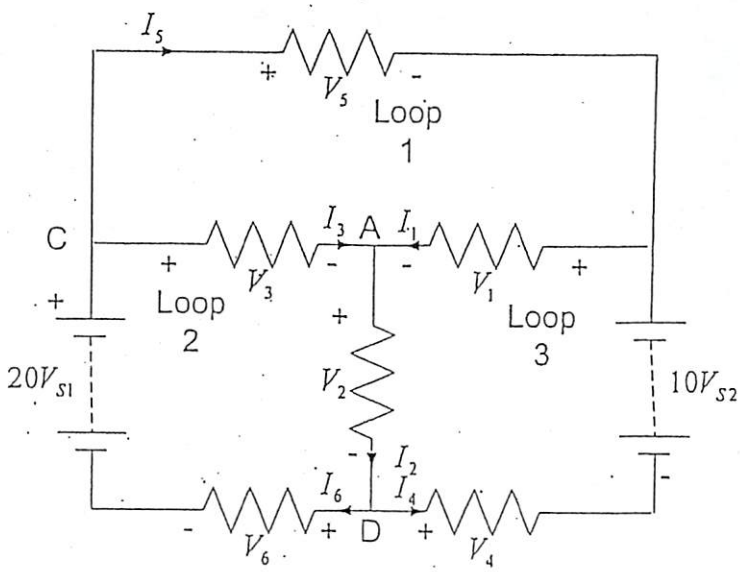
مركز خدمات كلية الهندسة
مبني ٣ الطابق الأرضي
غرفة رقم أ / أ/ ٢٩

تصوير
مذكرات
تعليف
قرطاسية
بطاقات

LABORATORY MANUAL FOR

EE 205:

ELECTRICAL CIRCUITS LABORATORY



مع تمنياتنا لكم بالنجاح

صندوق الطلاب

أهداف المعمل EE205

- إكتساب المهارة اليدوية.
- تطبيق المفاهيم النظرية بشكل علمي.
- الإلتزام بالوقت المحدد للحضور.
- الإلتزام بالوقت المحدد لإنهاء التجربة.
- العمل كالفريق .
- الإلتزام بتدابير السلامة.
- التعرف على طريقة التعامل مع الأجهزة الكهربائية كالبورد والأسلاك والبطاريات والمقاومات والمكثفات...الخ
- إكتساب المعرفة اللازمة لفهم الدوائر والخرائط الكهربائية.
- كتابة التقارير بطريقة علمية.
- التعامل مع مختلف الأجهزة الكهربائية.
- التعلم عن طريقة الملاحظة.
- إكتساب مهارة الثقة بالنفس.
- تحويل الكتابة إلى عملي.
- تحويل العملي إلى كتابة.

EE 205: EECTRIC CIRCUITS LABORATORY

Lab Report Marks:

#	Item	Marks
1	**Title Page** You must use the designed title page for each experiment. Do not use your own design.	2
2	**Objectives** Should be the same as given in the lab manual. If some objectives in the lab manual were not done in the laboratory, do not include them in your report.	2
3	**Theory or Background** You should write your own summary (not more than on page) of the theory, Do not copy directly from the lab manual or attach any part of it to your report.	2
4	**Experimental Work**	
	a) List of equipment used e.g. multimeters, oscilloscope, signal generator, etc.	2
	b) Circuit Diagrams: Must be computer drawn. Hand-drawn circuits are not acceptable.	2
	c) Procedure: Do not copy from manual. Write in the third person; avoid the use of the first person (I, we, our). For example, instead of saying: We connected the circuit in Fig.1....; say: The circuit in Fig.1 was connected ...	2
	d) Tables: Must have a title, grids, and units	3
	e) Calculations. Use equations, substitute, and calculate values. Must indicate units of calculated values. For example Ω, V, A, W, etc.	3
	f) Graphs: Must be drawn on standard graph paper or using computer graphics. Each graph must have a title. All axis must be properly labeled. The x-axis must represent the independent variable. If two y-axes are used, both must be use proper scales and labels.	3
5	**Comments and Conclusions** This is the most important part of your report, so read carefully from the lab manual for what you should write under this heading.	5
6	**Neatness and Clarity**	2
7	**Formatting and Sequence**	2
	Total Marks	30

Grading:

#	Item	Percentage
1	Attendance, Punctuality, and Behavior in the lab	10 %
2	Pre-Lab Quizzes	10 %
3	Lab Reports	30 %
4	Midterm Exam	20 %
5	Final Exam	30 %
	Total	100 %

Pre-lab Quizzes:

A short quiz will be given at the beginning of each experiment. Therefore, read the experiment before coming to the lab.

Reports:

A report about the previous experiment should be prepared and submitted to the teaching assistant before starting the new experiment.

Midterm Exam:

An examination will be given at the middle of the semester. This midterm examination will include some theoretical and practical concepts covered in the laboratory.

Final Exam:

The final examination is similar to the midterm exam, but will cover all the experiments.

Equipment:

Each team has to bring A4 white papers and commercial graph papers and a ruler. Calculators may be used during the quiz and the experiment. The students are responsible for the equipment on their experiment table. The equipment should be checked at the beginning of each laboratory session and in case of any missing equipment, the students should inform their assistants. Exchanging of equipment is not allowed among the tables without the permission of the teaching assistant.

Attendance:

The students should attend all the experiments. **Those who miss two experiments without any allowable excuse (medical report or coinciding exams) will fail the course.**

No student will be allowed to take lab in another lab day other than his own for any reason [other than a medical report from the health center of the university (**NO OTHER HEALTH CENTER!**) and any conflicting exam.

If you have a medical report or a conflicting exam, you should consult your teaching assistant with a written formal request and your request will be assessed whether you will be given a make- up.

Papers indicating that you were at the health center during your experiment time will not be counted as a medical report.

Course Announcements:

The students are responsible for regularly checking the KSU black-board for further announcements.

https://lms.ksu.edu.sa

HOW TO PREPARE A LABORATORY REPORT

Reporting the results obtained from an experiment is as important as conducting the experiment. A sloppy report significantly deducts from the value of the results. Therefore, a properly prepared report is very important.

Fortunately, in this course the lab report has a fairly consistent format that will help you to organize your information clearly. This document covers some important points on how to prepare a proper report for an EE205 experiment.

Cover Page

The cover page must include:

Section Number
Experiment Number
Experiment Title
Experiment Date
Report Date
Names and the ID-numbers of the Students
Name of Teaching Assistant

Sections of the Report

There are three main sections of the laboratory report.

Objective
Results
Conclusion

Objective:

In this section objectives of the report should be summarized. Laboratory manual also contains a section which explains the objectives of the experiment. The object section in the report should not be copied directly from this section. It should represent your understanding of the experiment with your own words.

Results:

This section contains the results obtained in the experiment. The results which consist of graphs should be given on separate graphical papers. You should compare your results with the expected results; in case they are different then you should state the possible reasons. It is strongly recommended that you enrich your results with your comments.

Conclusion:

The conclusion section is a verbal summary of the experiment. It is important that you do not repeat the results section here; this should be a simple summary of what was done, observed and learned in the experiment.

Plotting the Graphs

Some of the results are presented by graphs more efficiently. Therefore, graphs are frequently used in reports. A proper plot should obey the following rules:

- All of the graphs should be plotted on a graphical paper, or using computer graphics.
- Title of the graph should be written.
- The horizontal axis should be assigned to the independent variable
- The vertical and horizontal axes should be drawn and clearly named. The units of the axis should be written.
- Origin of the plot should be clearly indicated.
- The tick-marks corresponding to each unit on the axes should be depicted clearly.
- Units of the values should always be mentioned.
- Critical or important values should always be mentioned.
- If two variables are plotted on the same graph, it may be necessary to use two vertical axis with different scales, if one variable values are very small compared to the values of the other variable.

Example of a proper graph:

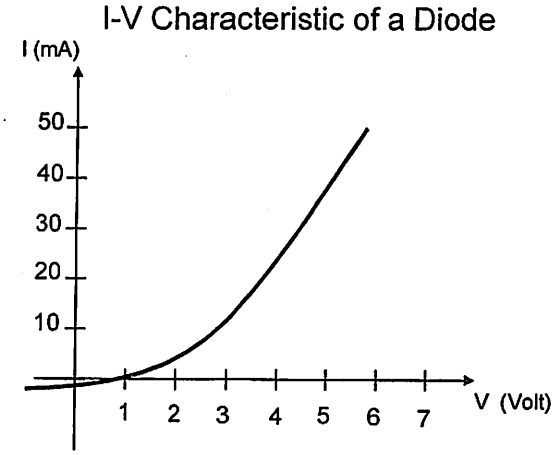


Fig.1: I-V Characteristics of a diode

An unacceptable graph:

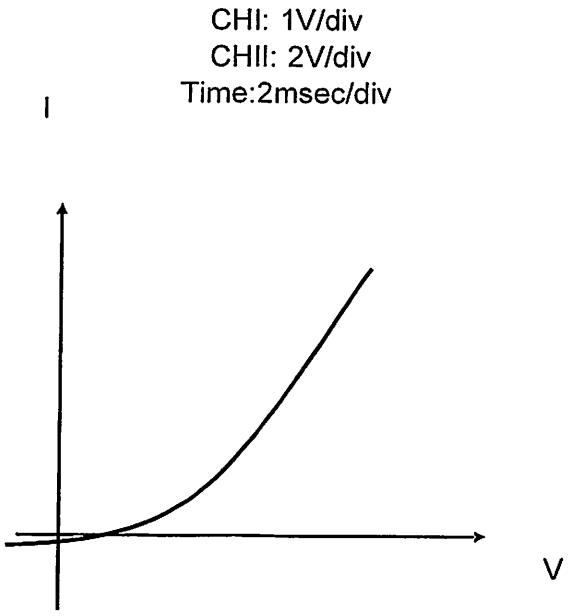


Fig. 2

The graph shown in Fig. 2 is **completely unacceptable** because,

1- The title of the graph is not written
2- Units of the axes are not given
3- Axes are not 'ticked'. Corresponding values are not shown.

King Saud University

College of Engineering

Electrical Engineering Department

EE – 205 Electrical Circuit Laboratory Experiment No. 1

Introduction to Measurement of Resistors, Current and Voltage Divider Rules, and Δ - Y Transformations Rule.

Objectives:-

1. Familiarization with the lab policy, electric circuit components, safety rules, and main equipment (multimeters, power supplies, decade resistance box).
2. Verification of voltage and current division rules.
3. Verification of Δ - Y Transformations.

Required Equipment:-

DC power supply.

Three color code resistors,

Three decade resistance box,

Digital Multimeters

220 Ω (two), 470 Ω, 1 KΩ, 2.2 K Ω, 4.7 K Ω, and 10 K Ω Resistors.

Background:-

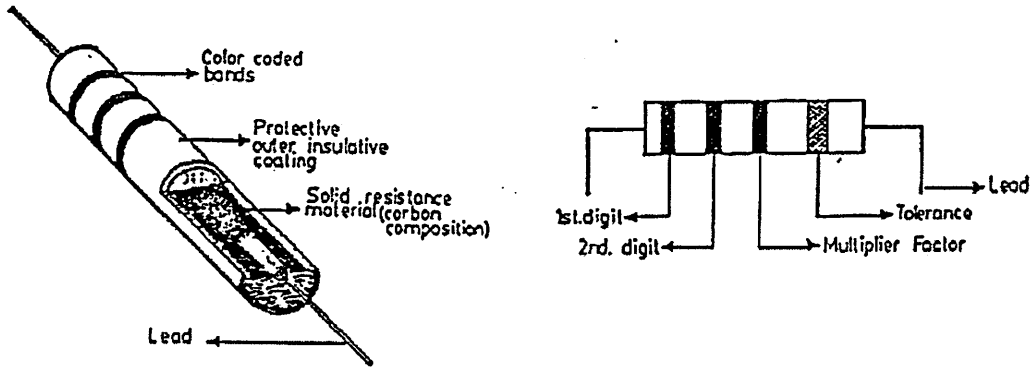
1. Resistors

Most discrete resistors fall into one of four basic categories: Carbon composition, carbon film, metal film, or wire wound. The most common of the low-wattage, fixed type resistors is the molded carbon composition resistor, basically shown in Fig.1-a. Carbon composition resistors have been used for nearly 100 years and are still popular. It is a mixture of carbon and an insulating material. The value of the resistance depends primarily on the ratio of the carbon to the insulating material. Practical resistors are manufactured in standard values, various resistance tolerances, various power (wattage) ratings, and in many forms of construction. The standard or nominal resistance values are listed in Table 1 for resistors from 1 to 9.1 Ω. Other values are available (from about 1Ω to about 22MΩ) in factors of 10. Thus resistors such as 8.2Ω, 82Ω, 820Ω etc are standard or nominal values. All values shown in Table 1 are available in a ± 5% tolerance (accuracy of resistance), but only certain values in Table 1 are available in other tolerances. For example, only the values in rows 1 and 3 are available also in ± 10% tolerance. For a ± 20% tolerance resistor, only the values in row 1 are usually available.

Table 1: Standard or nominal values of resistors having ± 5% resistance tolerance.

1.0	1.5	2.2	3.3	4.7	6.8
1.1	1.6	2.4	3.6	5.1	7.5
1.2	1.8	2.7	3.9	5.6	8.2
1.3	2.0	3.0	4.3	6.2	9.1

In many precision resistors of ± 1% accuracy or better, the resistance value is marked directly on the resistor. For the common carbon composition type illustrated in Fig. 1-a, standard color-coded bands or rings are used to define resistance as in Fig. 1-b. These colored bands indicate the resistance in ohms:



(a) Carbon composition resistor. (b) Resistor color code bands.

Fig. 1

The first digit is given by the band nearest the edge. The second digit is given by the next band. The number of naughts (zeros) [or the power n of the multiplying factor 10^n] is given by the third band.

The colors in these first three bands represent numbers as follows:

Color	Black	Brown	Red	Orange	Yellow	Green	Blue	Violet	Gray	White
Value	0	1	2	3	4	5	6	7	8	9

Gold in the third band means a multiplying factor of 10^{-1} while silver means 10^{-2}. The tolerance is indicated by the (fourth) band with a color code as: 1% brown, 2% red, 5% gold, 10% silver, and 20% black or no band.

2. Voltage Divider Rule

The voltage divider rule states that:

$$V_x = \frac{R_x}{R_x + R_y} E$$

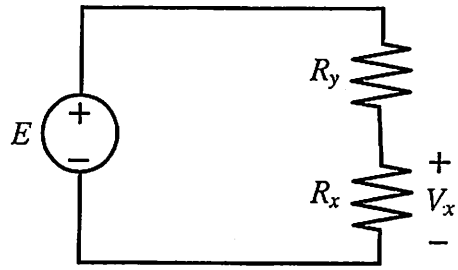


Fig. 2: Circuit diagram for Voltage division rule.

3: Current Divider Rule

The current divider rule states that:

$$I_x = \frac{R_y}{R_x + R_y} I$$

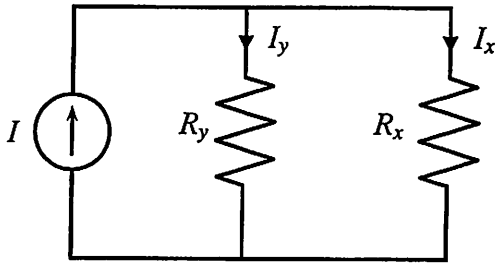


Fig. 3: Circuit diagram for Current divider rule.

4. Delta (Δ) and Wye (Y) Connections:

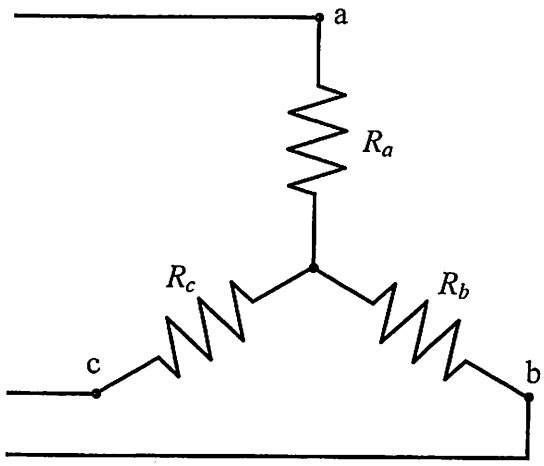
There are certain circuit configurations in which the resistors do not appear to be in series or in parallel. Under these conditions, it is necessary to convert the circuit under investigation from one form to another. Delta (Δ) and Wye (Y) Connections are the most common connections that exist in electric circuits. Δ-connection is also known as Pi (π)-connection and Y-connection is also known as star or Tee (T)-connection. To convert Δ to Y and vice versa, we use the following equations as applied to the notation of Fig. 4.

Δ to Y transformation or conversion

$$R_a = \frac{R_{ab} R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_b = \frac{R_{bc} R_{ab}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_c = \frac{R_{ca} R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$



Y-Δ transformation or conversion

Also one can write:

$$R_{ab} = R_a + R_b + \frac{R_a R_b}{R_c}$$

$$R_{bc} = R_b + R_c + \frac{R_b R_c}{R_a}$$

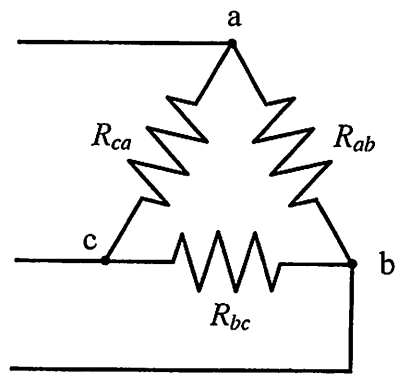


Fig. 4: Δ - Y Transformations.

$$R_{ca} = R_c + R_a + \frac{R_c R_a}{R_b}$$

Procedure I:-

Resistance Measurement:

1. Given the three color coded resistors found on your experiment bench, determine the value of each first by calculation using the color code and second by measurement with an ohmmeter. Record your readings below and comment on the differences.

Table 2: Resistance measurement by color code and by ohmmeter.

Resistor	Colors	Value by Calculation	Value by Measurement
1			
2			
3			

Comment:

2. Using an ohmmeter, measure the assigned different resistances from the decade resistance box as shown in table 3, record your readings. Also calculate the tolerance as shown on the decade resistance box and the errors of your readings.

Table 3: Resistance measurement and calculation of tolerance and errors.

Readings	Decade Box Range	Assigned Resistance	Ohmmeter Reading	% Tolerance	% Error
1.	× 1	4 Ω			
2.	× 10	50 Ω			
3.	× 100	700 Ω			
4.	× 1 K	5 K Ω			
5.	× 10 K	30 K Ω			

(a) Percentage Tolerance:

3. Using the percentage tolerance errors, indicated on the box ranges, calculate the acceptable ranges of each resistor value, and record your result in table 2 above.

Example:

Range ×1 has % tolerance of 0.5%, then the resistance value of 4 Ω has a tolerance of 4× (0.5/100) = ± 0.02 Ω and the real resistance value is located within 4 ± 0.02 Ω i.e. 3.98 Ω to 4.02 Ω.

(b) Percentage Error:

4. Calculate the % error of your readings (measurements) for each meter according to the following:

$$\%\ error = \frac{Actual\ meter\ reading - \text{standard}\ (decade\ box)\ reading}{\text{standard}\ (decade\ box)\ reading} \times 100$$

Procedure II:-

Voltage Divider Rule:

1. Connect the circuit shown in Fig. 5.
2. Measure V_1, V_2, V_3, and E accurately.
3. Calculate V_1, V_2, and V_3 using the voltage division rule.
4. Compare the values of V_1, V_2, and V_3 in steps (2), (3) and (4).
5. Record your results in Table 4.
6. Comment on the differences (if any) between values V_1, V_2, and V_3 in steps (2), (3), and (4).

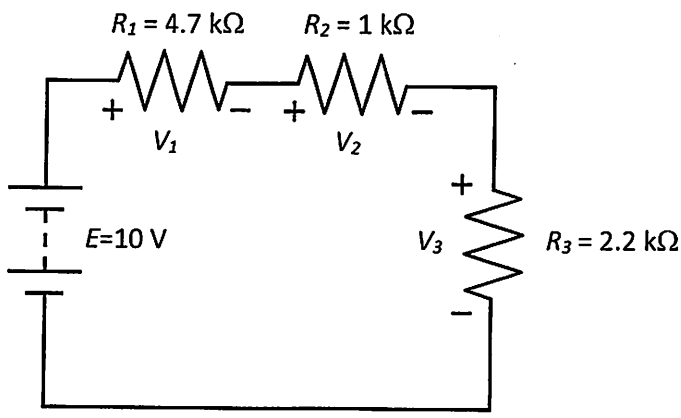


Fig. 5: Circuit to verify voltage division rule

Table 4: Results for verification of voltage division rule.

E = 10 Volts

	V_1 (V)	V_2 (V)	V_3 (V)	V_{Total} (V)
By Measurement				
By Calculation				
% Error				

Procedure III:-

Current Divider Rule:

1. Connect the circuit shown in Fig. 6.
2. Measure I_1, I_2, I_3, and I_T accurately.
3. Calculate I_1, I_2, I_3 using the current division rule.
4. Record your results in table 5.
5. Comment on the differences (if any) between values I_1, I_2, I_{3m} and IT in steps (2), (3), and (4).

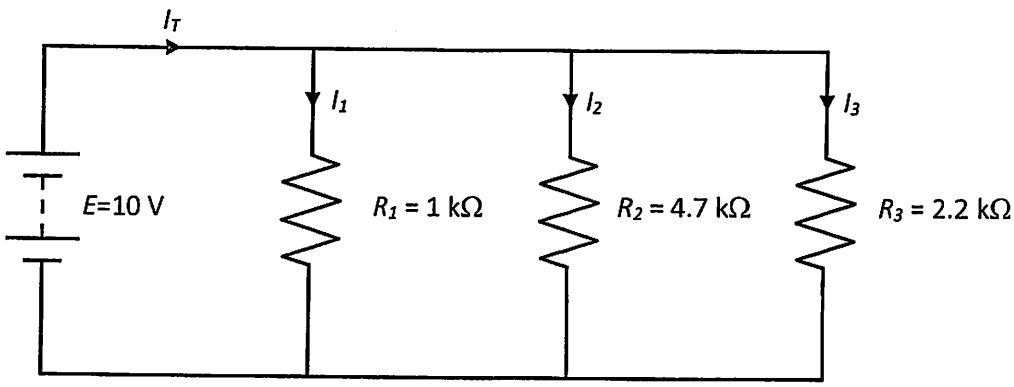


Fig. 6: Circuit to verify current division rule.

Table 5: Results for verification of current divider rule.

	I_1 (mA)	I_2 (mA)	I_3 (mA)	I_T (mA)
By Measurement				
By Calculation				
% Errors				

Procedure IV:

Δ-Y Transformation:

1. Connect the circuit in Fig. 7 and calculate the current I_D using any method **except Δ-Y transformation.**

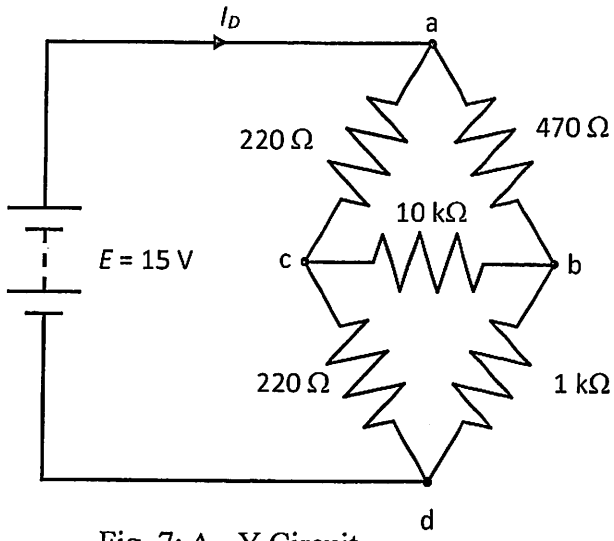


Fig. 7: Δ - Y Circuit.

2. Measure I_D accurately.
3. Compare the calculated, measured, and simulated values of I_D.
4. Calculate the equivalent Y for Δ abc and record the values of the resistors *Ra*, *Rb*, and *Rc*.
5. Connect the circuit of Fig. 8 and measure I_Y.
6. Compare the measured, calculate, and simulated values of I_D and I_Y

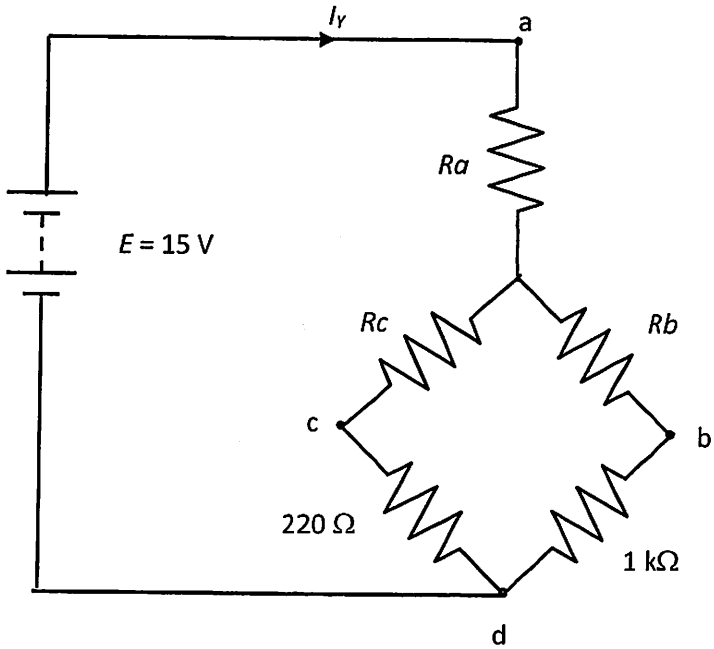


Fig. 8: Δ - Y Circuit.

King Saud University
College of Engineering

Electrical Engineering Department

EE-205: Electrical Circuit Lab Experiment No. 2

Root-Mean-Square (RMS) Value, Peak, and Peak-to-Peak Values, Measurements with Oscilloscope

Objectives

Investigation of:

- The relationship between rms and peak-to-peak values of a sinusoidal wave.
- Use of Oscilloscope in measuring different parameters and evaluation of Phase difference, Angle, Time Period and Frequency under sinusoidal excitation.

Required Equipment

Resistors:	100 Ω and decade box
Inductors:	17.5 mH, 1000 tuns air core
Equipment:	DC supply, Signal Generator, Digital Multi-Meter (DMM), and Oscilloscope

Background

I RMS, Peak, And peak-to Peak Values

As we studied earlier, the DC current and voltage do not change the direction and polarity, respectively; they maintain a constant output level for a particular load. However, alternating current and voltage change continuously. Moreover, the polarity of the ac voltage, and the direction of ac current, change periodically. The reason of using ac voltage and current in modern power systems is that they are easy to generate and less expensive to transmit to users.

The RMS (Root-Mean-Square) value of an ac voltage or current x(t) is defined by the relation:

$$x_{rms} = \sqrt{\frac{1}{T}\int_o^T x^2(t)dt}$$ where T is the period of the ac wave

Note: The multimeters used in this lab are, generally, rectifying-type meter. It does not indicate correctly the rms value for non-sinusoidal waveforms.

It can be shown for a sine wave:

$$x_{rms} = \frac{1}{2\sqrt{2}} \text{ x (peak-to-peak value)} = \frac{1}{\sqrt{2}} \text{ x (peak value)}$$

$$\text{or } x_{rms} = \frac{1}{2\sqrt{2}} X_{p.p} = \frac{1}{\sqrt{2}} x_p$$

II The Oscilloscope

The oscilloscope is a new instrument to be used from now on in the experiments. Essentially it is a voltage detecting instrument. It allows you to see the variations of a voltage or current with time. All of the controls for the proper operation of an oscilloscope are mounted on the front panel of the instrument.

The magnitude of a given voltage using an oscilloscope is given by:

Volts (unknown) = Voltage sensitivity (volts/division) × Deflection (vertical divisions)

The current measurement is done by measuring the voltage across a resistor and dividing by the value of that used resistor. The used resistor should be much less than the total resistance of the circuit.

III AC Circuit analysis

Analyzing ac circuits in time domain (real analysis) is a difficult task. So different methods have been introduced to simplify the analysis; among these is what is called phasor algebra (complex analysis). The different methods of circuits analysis as KVL, KCL, voltage division rule, current

division rule, superposition, Thevenin's Theorem, Norton's Theorem, Maximum Power Transfer Theorem, and others, are still applied to analyze ac circuits using the phasor vector analysis.

Important Rules to Remember:

$\overline{Z}_L = j\omega L = \omega L \angle 90^D$= impedance of the inductor in phasor form

$\overline{Z}_C = 1/j\omega C = (1/\omega C)\angle 90^D$= impedance of the capacitor in phasor form

$\overline{Z}_R = R = R\angle 0^D$= impedance of the resistor in phasor form

$\overline{Z} = Z_1 + Z_2 + \ldots\ldots$ total impedance in series ac circuit

$F_P = \cos\theta_T = P/EI = R/Z_T$

F_P : power factor
P : average power delivered (active)
θ_T : the angle of impedance vector
E : magnitude of the supply voltage
I : magnitude of current

$P = EI\cos\theta_T = I^2R$

From impedance diagram:

$Z_T = \sqrt{R^2 + (X_L - X_C)^2} =$ magnitude of the impedance of the circuit.

$\theta_{ZT} = \tan^{-1}(X_L - X_C)/R$= angle of the impedance

Where $X_L = \omega L$= reactance of the inductance L
$X_C = 1/\omega C$ = reactance of the capacitor C

$$E = \sqrt{V_R^2 + (V_L - V_C)^2}$$

Where V_R = magnitude of the voltage across the resistance R
V_L = magnitude of the voltage across the inductance L
V_C = magnitude of the voltage across the capacitance C

Procedure I:

<u>Familiarization with the Oscilloscope</u>

1. Turn on the oscilloscope and observe the function of the following control switches.
   a) Focus b) Intensity c) Horizontal position, and vertical position
2. Connect the circuit shown in Fig. 1.

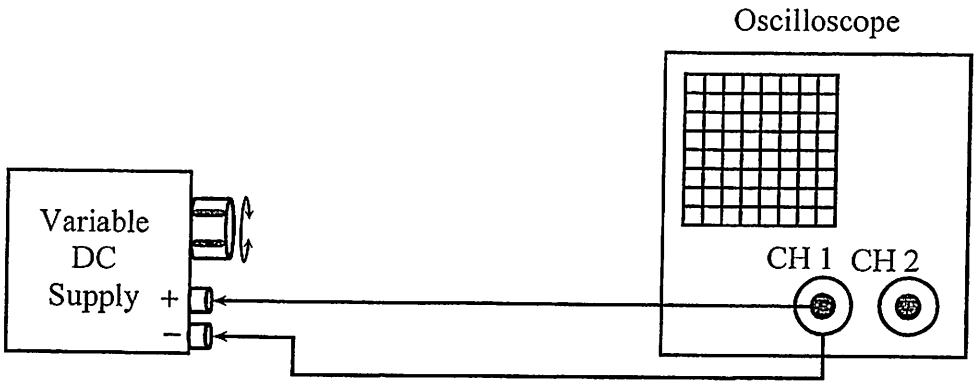


Fig.1: Oscilloscope for measurement of DC Voltage.

a) Locate the zero line on the scope screen.
b) Use channel-1 switch to be at dc mode
c) Vary the supply voltage to obtain the following values recorded in the table, and fill up the table with necessary information:

DC Supply Voltage	Vertical scope deflection (divisions) × vertical sensitivity (V/div) = measured voltage by Oscilloscope
1 V.	
2 V.	
5 V.	
10 V.	

3. Turn the dc voltage supply off and keep your circuit connected.
4. Switch the display mode switch to channel 2 and wire the circuit shown in Fig.2. Use the function generator as an AC supply.
5. The channel 2 input switch should be set on ac mode. Display a sinusoidal signal on the Oscilloscope and calculate its frequency and magnitude as follows:

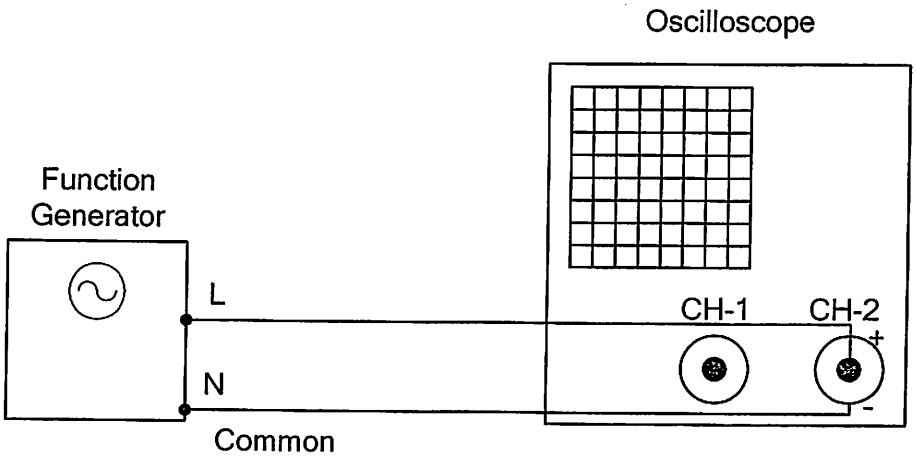


Fig. 2: Oscilloscope for measurement of AC voltage.

Ac signal	Peak voltage (V) = vertical scope Deflection (divs.) × vertical Sensitivity (V/div) Vm	Period (ms) = Horizontal Scope deflection (divs) × Horizontal sensitivity (ms/div) T	f = 1/period
Sinusoidal Waveform			

6. Calculate the frequency of the displayed signal $f = 1/T$, and the V_{rms}
7. Use the Multi-meter (ac voltage mode) to measure the V_{rms} of the signal. How does this value compare to the calculated value in (6)?
8. Return the dc supply ON for the circuit of Fig.1. Connect to channel 1 and switch the display mode switch to DUAL (which means that the two displayed signals will be traced simultaneously on the scope). Then switch the mode to (ADD) mode (which means that the displayed signal will be the addition of signals entering channels 1 and 2 simultaneously). How does the displayed signal compare to the total (addition) of signal of 10 V dc and the signal of the frequency and magnitude described in part (5)? What would be the V_{rms} for the new signal?

Procedure II:

Comparison Between Oscilloscope and Digital Multi-Meter for RMS Measurement

1- Use the function generator to generate a sinusoidal voltage of 10-V (peak to peak).
2- Use the oscilloscope to measure the peak to peak value (10 V). RMS = 5/(sqrt(2)= 3.535V
3- Use the digital multi-meter to measure the RMS value.
4- Increase the frequency (keeping peak to peak value constant) and record RMS value in as follows:-

V (peak to peak)	Frequency (Hz)	RMS Value Using Oscilloscope	RMS Value Using Digial Multimeter	% Error
10 V	100 Hz			
10 V	1000 Hz			
10 V	2000 Hz			
10 V	3000 Hz			
10 V	4000 Hz			
10 V	5000 Hz			

Procedure III:

Series RL Circuits Measurements

1. Measure the resistance of the coil, R_L.
2. Connect the circuit shown in Fig. 3 and measure E, V_L, and V_R using a Multi-meter.
   Note: The measured values obtained are the rms values.
3. Display V_R and E on the Oscilloscope. Plot the waveforms.
4. Measure the phase angle θ, and indicate the leading waveform. Explain why?
5. Remove the Oscilloscope connection and measure the current I flowing in the circuit.
6. Determine Z_T using the relation: $Z_T = E/I$.
7. Calculate Z_T and draw the impedance diagram used in your calculation.
8. Using phasor analysis prove that $E = V_R + V_L$
9. Calculate the average power delivered and the power factor for the circuit of Fig.3.

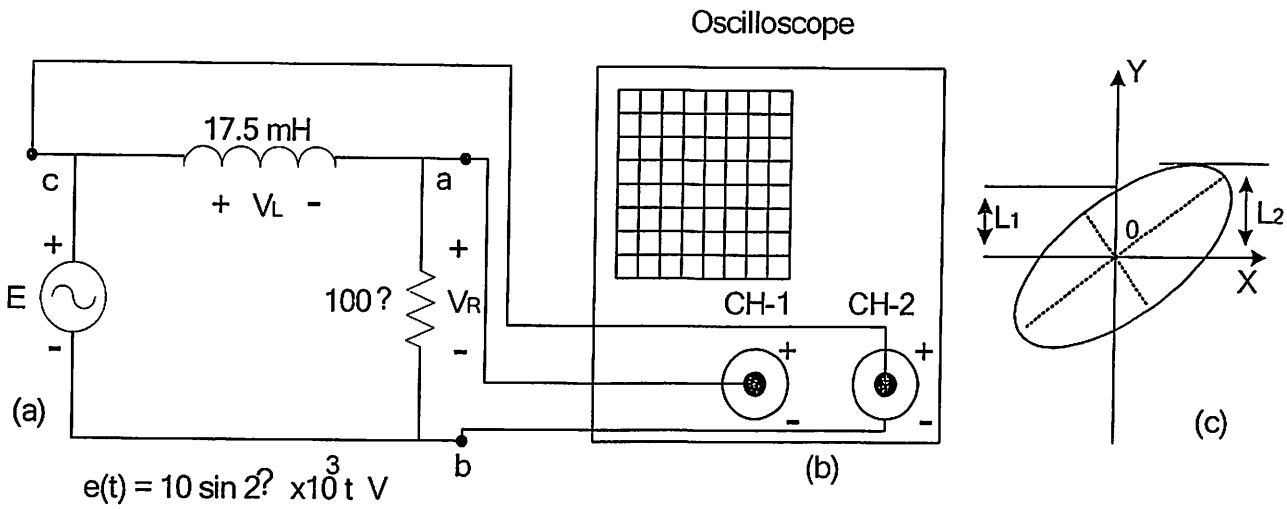


Fig. 3: Circuit connection for procedure II.

Lissajous pattern on the oscilloscope can be used to measure the phase angle between two equal-amplitude deflection signals defined as:

$\theta = \sin^{-1}(L_1/L_2)$; see Fig.3-c

Note: You would display the above pattern by using X-Y mode of the oscilloscope.

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EE-205 Electrical Circuit Lab Experiment No. 3

Thevenin's and Maximum Power Transfer Theorems

Objectives

- To verify experimentally Thevenin's and maximum power transfer theorems as applied to ac circuits.

Required Equipment

Resistors: 100 Ω, 1 kΩ, 3.3 kΩ, 10 kΩ, one decade resistance box.

Inductors: 17.5 mH inductor with iron core.

Capacitors: Two 0.1 μF, 0.47 μF, 1 μF, 4.7 μF, 22 μF.

Equipment: Multimeter, Signal generator.

Thevenin Theorm

Thevenin's theorem can be applied to ac circuits in the same manner as has been applied to dc circuits. To find the variable of interest, isolate the related branch from the circuit, short out all the independent voltage sources and open all the independent current sources (this can be done only in theory, not in practice) and calculate Z_{TH} across the terminals in question. Then replace all sources and calculate E_{TH}, which is the voltage across the terminals of question. Draw the Thevenin's equivalent and replace the branch that has been isolated and find using basic rules the desired variable (voltage, or current etc.). See Fig.1

22μ
1 kΩ
100 Ω
10 kΩ

22μ
10kΩ
1K

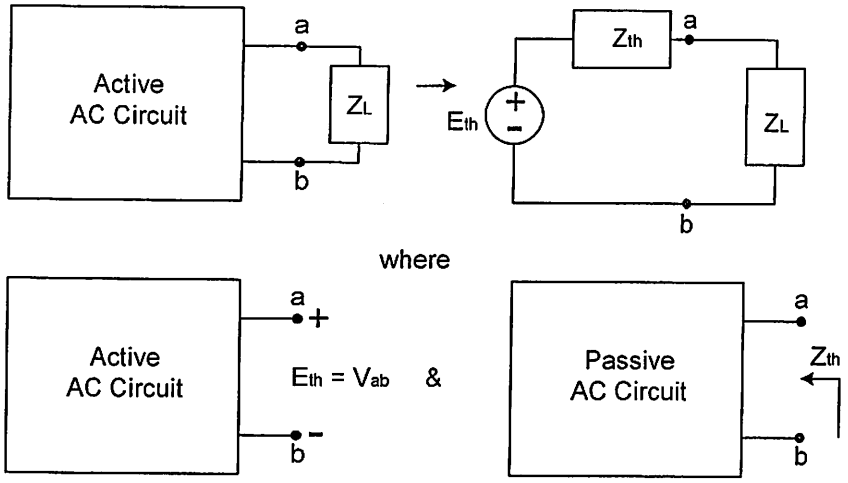


Fig. 1: Thevenin's Theorem.

II. Maximum Power Transfer Theorem

Maximum power transfer theorem in ac circuits states that the max power will be transferred to a load when the load impedance is the conjugate of the Thevenin's impedance across the load terminals. See Fig.2

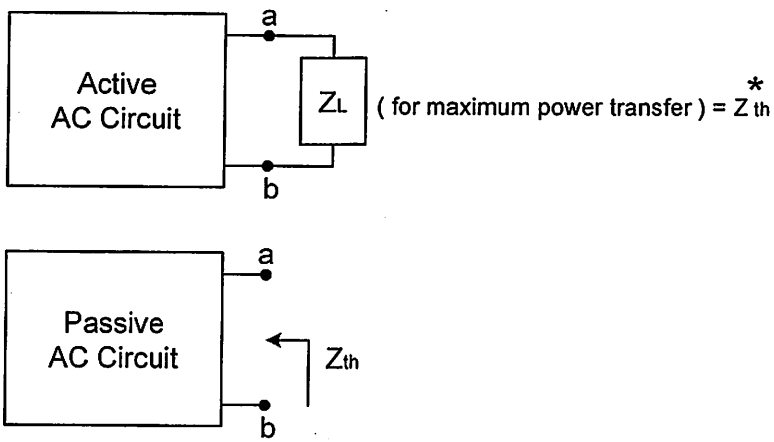


Fig. 2 Maximum power transfer theorem.

Procedure I:

1. Connect the circuit shown in Fig. 6a, and measure V_R and I_R.
2. Isolate the (10 KΩ) resistor from the circuit and measure the voltage across the terminals (a-b). This voltage is E_{TH}.
3. Calculate Z_{TH} by shorting the supply.
4. Reconstruct the Thevenin's equivalent circuit for the circuit of Fig.6-a by inserting the E_{TH} and Z_{TH} as shown in Fig.6-b and measure I_R. and V_R. Compare this with step (1).

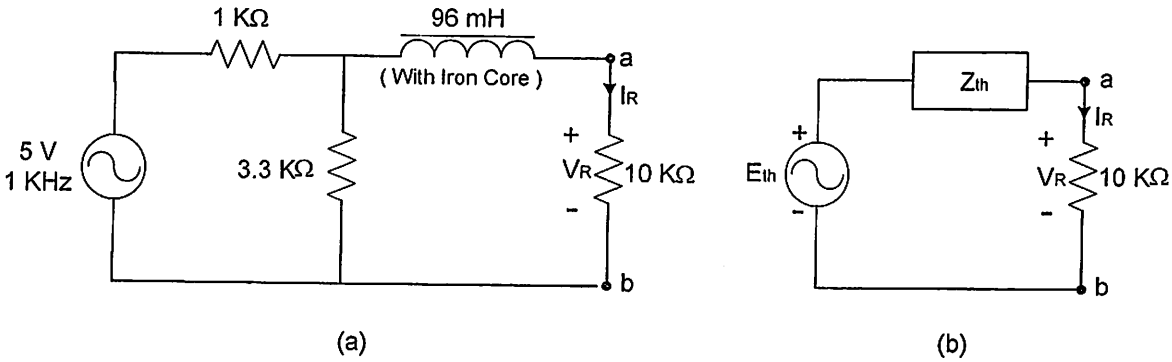


Fig. 6: Circuits for procedure I.

Procedure II :

1. Construct the circuit of Fig.7. Set $R_D = R + R_L$

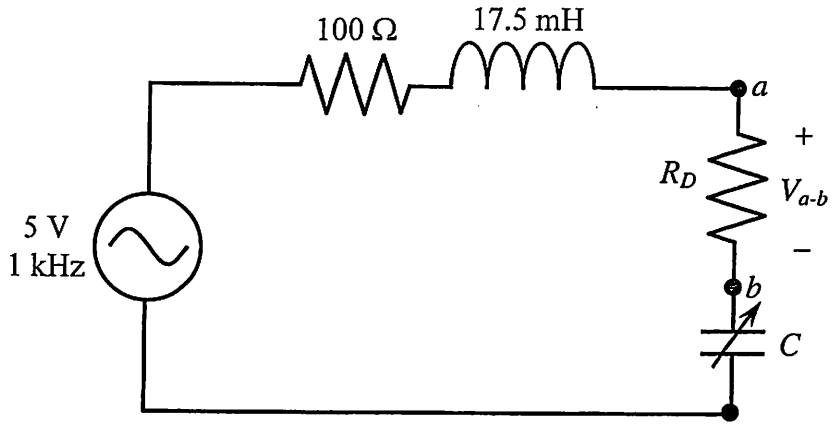


Fig.7: Circuit for Procedure II

2. Replace different capacitor combinations as shown in Table 1 and measure V_{a-b} for each combination.
3. Calculate Z_{TH} for the circuit of Fig. 7 and record it in the *rectangular* form.
4. Calculate X_C for each combination of capacitor and list the data in Table 1.
5. Calculate the power transferred to the load using:

$$P = \frac{V_{a-b}^2}{R_D}$$

Repeat this for all combinations of capacitors and list this data in Table 1.

6. On a graph, plot P vs $|X_C|$ and locate the maximum power transfer on the graph what can you conclude about the maximum power transfer?

Table 1

Combination of capacitors	$V_{a\text{-}b}$ (V)	$\lvert X_C \rvert$ (Ω)	P (mW)
0.1 μF			
0.47 μF			
0.57 μF			
1 μF			
1.1 μF			
1.2 μF			
1.47 μF			
4.7 μF			
22 μF			

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EE-205 Electrical Circuit Lab Experiment No. 4

Power Factor Improvement

Objectives

To introduce the concept of power factor and method of power factor improvement for a single phase inductive load.

Required Equipment

One Fluorescent lamp. (110 V, 20 W)

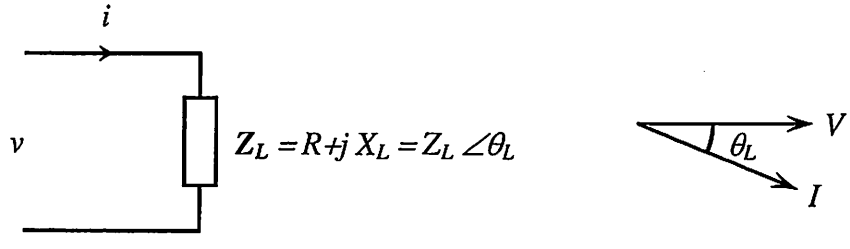
Two multimeters.

One Wattmeter.

One decade capacitor box.

Background

Consider a single phase circuit as shown in Fig.1 (a)



(a) (b)

Fig.1: (a) circuit. (b) phasor diagram.

If $v = \sqrt{2}\ V \sin\omega t$, the current $i = \sqrt{2}\ I \sin(\omega t - \theta_L)$, where $I = V/Z_L$, is the rms value of current and θ_L is its phase angle. The instantaneous power $P = vi = v = 2\ VI \sin\omega t \ \sin(\omega t - \theta_L)$ can be expressed as

$$P = VI \cos\theta_L - VI \cos(2\omega t - \theta_L) \qquad (1)$$

In equation (1), the first term is constant while the second term is a cosine wave which has a frequency of 2ω and a peak value of *VI*. The average value of this term is zero. Thus the average value of power delivered to load, P_{av}, is

$$P_{av} = VI \cos\theta_L = VI \ PF \qquad (2)$$

The term cos θ_L in equation (2) is called power factor (PF) of load. If the PF of a load is less than unity it may be either capacitive (i.e. leading) PF or inductive (i.e. lagging) PF. However, in practice almost all the loads are *inductive with lagging PF*. When the PF of a load less than unity, it takes more current for the same power from the supply. This large current requirement will cause more I^2R losses and *IR* drop in the supply line. Moreover, large current will demand greater current capacity for the generator and associated systems. Therefore, it is desired to correct the PF to unity. Lagging PF can be improved by connecting capacitors across the inductive load as shown in Fig.2. The shunt capacitor will not absorb real power but instead it will take leading current from the line to neutralize the lagging current component of the load, thereby making the net supply current less and hence improving the system PF.

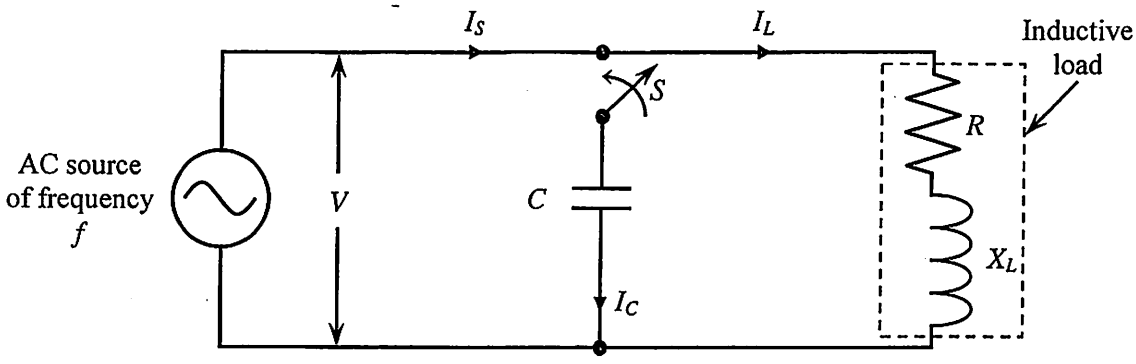


Fig.2: Inductive load with compensating capacitor.

For the circuit of Fig.2, load current I_L is given as

$$I_L = I_S = \frac{V}{R + jX_L} = \frac{RV}{Z_L^2} - \frac{jVX_L}{Z_L^2} \qquad (3)$$

where $Z_L^2 = R^2 + X_L^2$. The first component of the current in equation (3) (i.e. $R\ V/Z_L^2$) is called active component while the second component in equation (3) is called reactive component of

current. This reactive component which is lagging the supply voltage by 90° can be neutralized if a capacitor is connected across the load which draws a reactive current of similar magnitude but leading the supply voltage by 90° (i.e. $Ic = j\,V X_L / Z_L^2$). Thus, with the capacitor connected parallel to the load, the supply current Is is given as

$$I_S = I_L + I_C = \frac{RV}{Z_L^2} - \frac{jVX_L}{Z_L^2} + \frac{jVX_L}{Z_L^2}$$

or

$$I_S = \frac{RV}{Z_L^2} \qquad (4)$$

since . $Ic = j\,V/Xc = \mathrm{j}\,V(2\pi f C)$. Thus, to get a PF of unity, we must put C such that

$$V(2\pi f C) = \frac{V X_L}{Z_L^2}$$

or

$$C = \frac{X_L}{2\pi f Z_L^2} = \frac{X_L}{\omega Z_L^2} \qquad (5)$$

If C is less or more than that given by equation (5), the resulting power factor will be less than unity and will be lagging or leading respectively.

Fig.3(a) depicts the phasor diagram of voltage and current whereas Fig.3(b) shows the power diagram. Both of these diagrams show how the leading current or leading VA of the capacitor neutralizes the lagging reactive current or lagging reactive voltampere of the load, thereby, improving the PF.

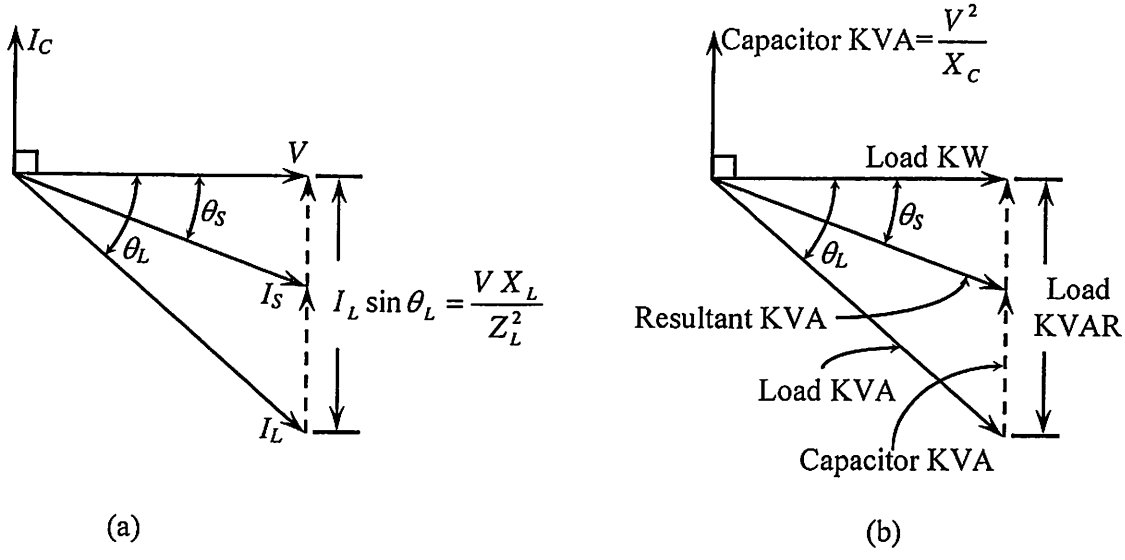


Fig.3: phasor diagram (a) Voltage, current, phasor diagram, (b) Power diagram

In both of these diagrams we see that initial or load phase angle, θ_L (before compensation) is more than the final or supply phase angle, θ_S (after compensation), i.e.

$$\theta_S < \theta_L$$

$$\cos\theta_S > \cos\theta_L$$

Procedure:

1. Connect the circuit as shown in Fig.4. *Get the connection checked by the instructor and let him energize the circuit for you.*
2. Record the reading of voltmeter, ammeter and wattmeter before connecting the capacitor.
3. Connect the capacitor to the line with a capacitance value of 1 μF, and record the reading of voltmeter, ammeter and wattmeter again.
4. Increase the value of the capacitor in steps and record the readings until you get a point where the current is minimum.
5. for the fluorescent lamp, without capacitor, estimate the load resistance R, inductive reactance X_L, impedance Z_L, phase angle θ_L and PF. Calculate the value of C required to make PF = 1.
6. Calculate PF for each value of C and complete Table 1. Find the value of C which give the highest PF. Compare these value with that calculated in step e.

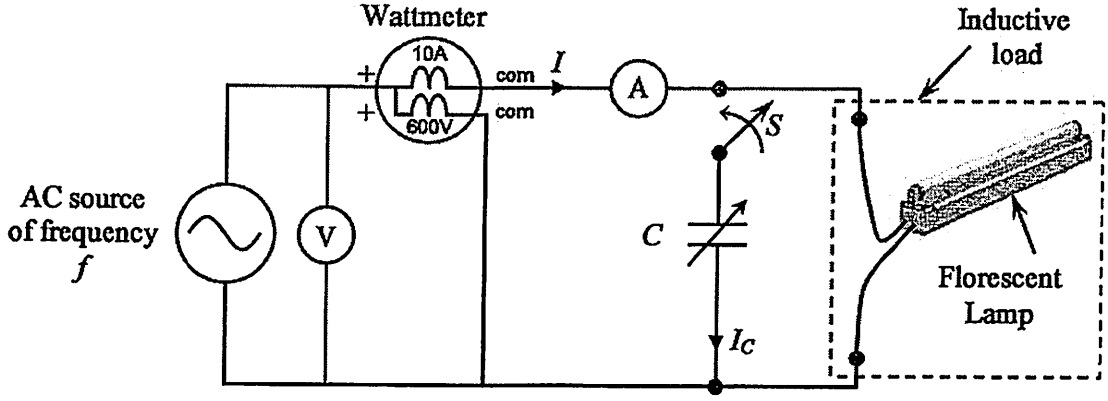


Fig.4: Experimental set up.

Table 1: Readings of Voltmeter, Ammeter, and Wattmeter with different value of C.

C (μF)	V (V)	I (A)	P (W)	PF
C *not connected*				
1.0				
2.0				
3.0				
3.2				

3.4				
3.6				
3.8				
4.0				
4.2				
4.4				
4.6				
4.8				
5.0				
5.2				
5.4				
5.6				
5.8				
6.0				

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EE-205: Electrical Circuit Lab Experiment No. 5

Three Phase AC Circuits

Objectives

- To get acquainted with the different methods of 3-phase connections and the corresponding relations between voltages and currents.
- To study balanced 3-phase circuits.
- To measure 3-phase power.

Required Equipment

Resistors: Three similar loading resistors

Equipment: Three ac ammeters, two ac voltmeters, two wattmeters and three-phase transformer.

Background

I. General

Most of electrical energy is generated and transmitted using 3-phase system. Three-phase ac supply system have the following advantages:

1. It is so far the most economical system from the point of view of generation, transmission and distribution.
2. It can feed electrical energy to a wide variety of loads. For example the loads may be three-phase loads or combinations of single and/or three-phase loads.

On the utilization side, three phase motors enjoy a uniform torque, while single phase motors do not.

II. The Three-Phase Power Supply

The following points are to be noted for the three phase power supply.

1. The three-phase supply is usually obtained from a three-phase ac generator having three symmetrical windings as shown in Fig.1. The magnitude axis of the winding of each phase is shifted from that of the other by 120°.

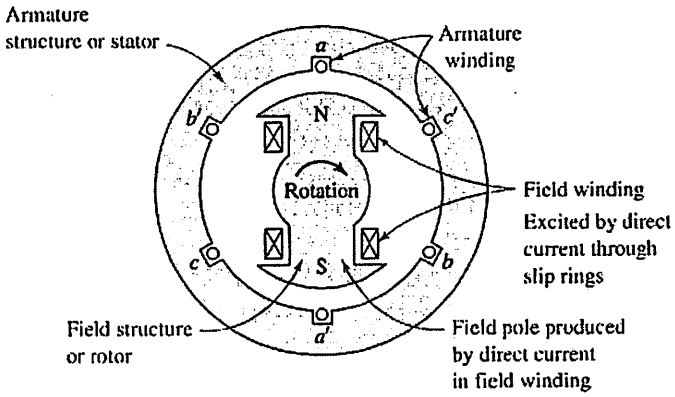
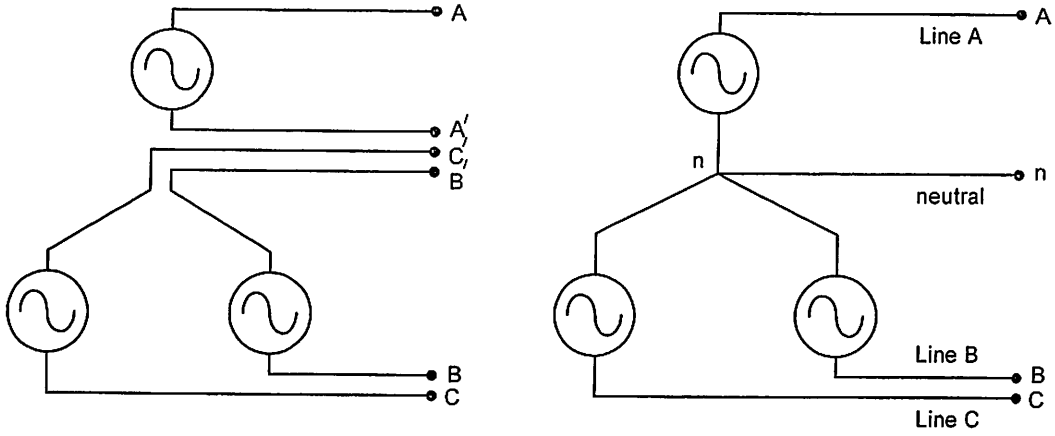


Fig. 1: A simple 3-phase generator with a 2-pole rotor

2. The three-phase supply is equivalent to three single phase generators as shown in Fig.2. Points A, B, C which represent the ends of each generator phase winding are usually connected together to form one point, n, known as the neutral.



(a) Three single-phase supplies. (b) a Y-connected three-phase supply.

Fig. 2 : A three-phase supply.

3. In the normal case of operation, the following time relations can be written:

$$V_A = V_{An} = V_{AA'} = V_{Max} \cos \omega t = \sqrt{2} V_p \cos \omega t$$

$$V_B = V_{Bn} = V_{BB'} = V_{Max} (\cos \omega t - 120^\circ) = \sqrt{2} V_p (\cos \omega t - 120^\circ)$$

$$V_C = V_{Cn} = V_{CC'} = V_{Max} (\cos \omega t + 120^\circ) = \sqrt{2} V_p (\cos \omega t + 120^\circ)$$

It means that the three phase voltages are equal in magnitude but shifted in phase by 120°. The supply in this case is known as a balanced Y-connected three phase supply.

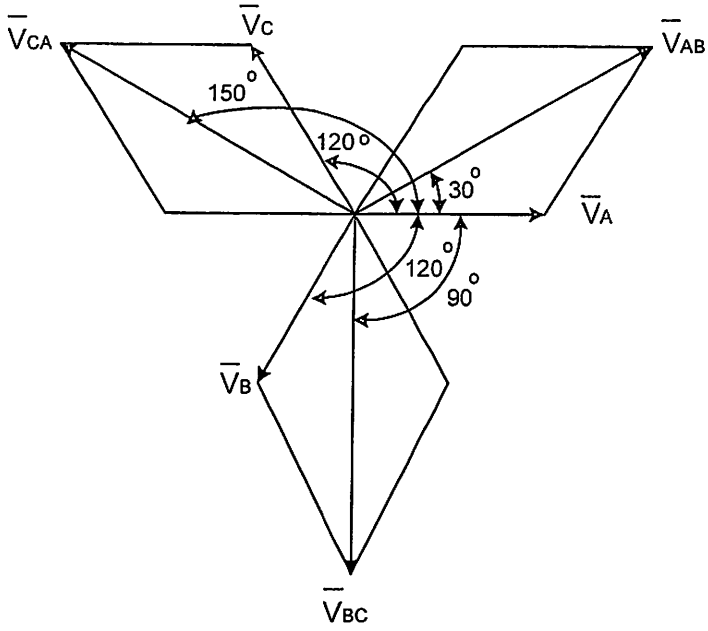


Fig. 3: Phasor diagram of the three-phase balanced supply.

Here, three new voltage phasors namely V_{AB}, V_{BC} and V_{CA} are introduced. These are known as line-to-line voltage phasors or simply line voltage phasors.

The time equations of these line voltages are:

$$V_{AB} = \sqrt{2} V_l \cos(\omega t + 30^\circ)$$

$$V_{BC} = \sqrt{2} V_l \cos(\omega t - 90^\circ)$$

$$V_{CA} = \sqrt{2} V_l \cos(\omega t + 150^\circ)$$

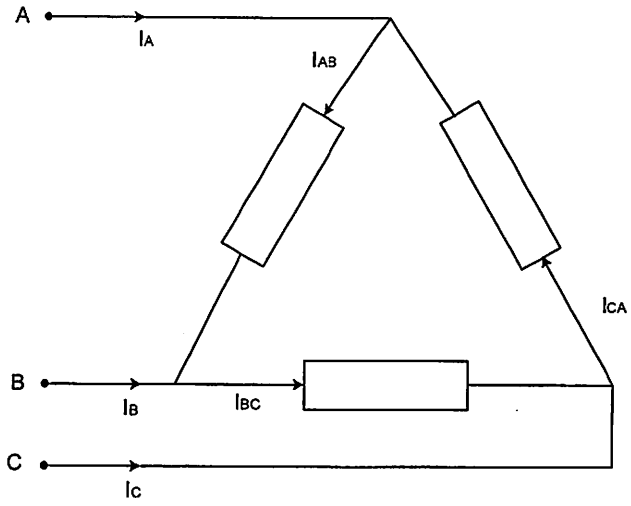
The magnitude of the line-to-line voltage V_l is $\sqrt{3}$ times the magnitude of the phase voltage V_p i,e:

$$V_l = \sqrt{3} V_p$$

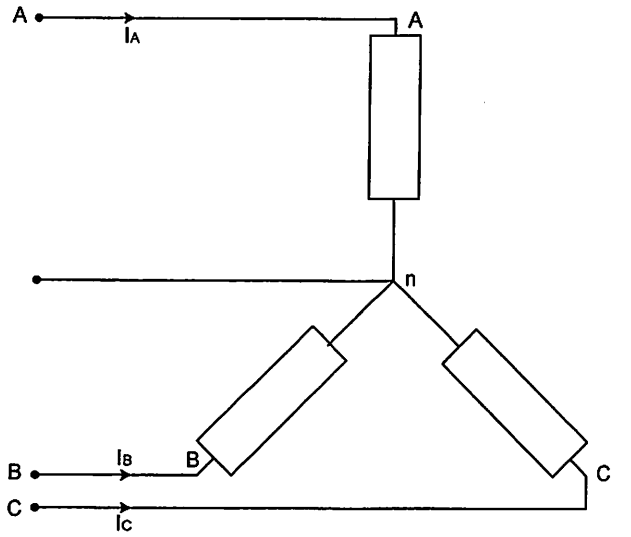
There is also a phase shift between the line and the phase voltage phasors. It is worthy to note that both phase voltages V_A, V_B, V_C and line voltages V_{AB}, V_{BC} and V_{CA} form a balanced three phase set.

III. Three-Phase Balanced Loads

A three-phase supply system delivers power to loads. The loads may be connected in star (wye) or delta shape of connections. Such schemes of connections are shown in Fig.4 for the case of balanced load (i,e. individual impedances are equal). The relations between the phase quantities and line quantities as well as the expression for the power consumed are given in the following:



(a) Delta connected balanced load.



(b) Y-connected balanced load with neutral connected.

Fig. 4: Three-phase loads.

a) for the □□connected balanced load

$$I_l = \sqrt{3} I_p$$

$$V_l = V_p$$

$$P = 3 V_p I_p \cos\Phi$$

$$= \sqrt{3} V_l I_l \cos\Phi$$

b) For the Y-connected balanced load:

$$I_l = I_p$$

$$V_l = \sqrt{3} V_p$$

$$P = 3 V_p I_p \cos\Phi$$

$$= \sqrt{3} V_l I_l \cos\Phi$$

where I_l and I_p are the effective or rms values of the line current and phase current, respectively. V_l and Vp are the effective or rms values of the line voltage and phase voltage, respectively. Φ is the phase angle between the current in each phase and the voltage across it, i,e. between V_p and I_p.

Note: The above relations are not generally valid for unbalanced three phase loads or generators.

IV. Three-Phase Power Measurements

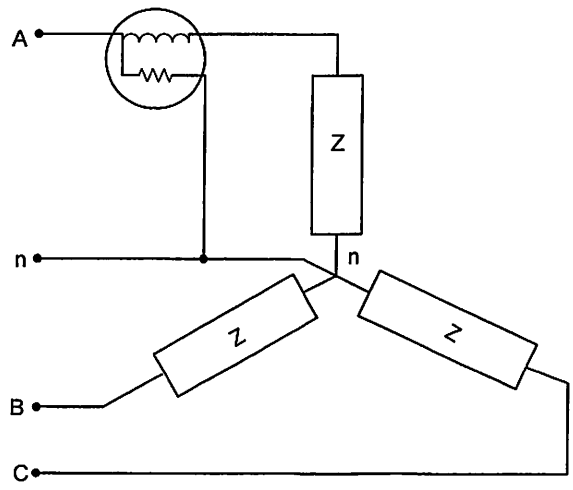


Fig. 5: Measuring the power of a balanced three-phase circuit using a single wattmeter.

For a balanced three-phase circuit, it is sufficient to measure the power in only one phase using a single wattmeter as shown in Fig.5. The total power is then three times the reading of this wattmeter.

For an unbalanced three-phase load, one wattmeter is not sufficient and the need for using three wattmeters , one for each phase, arises. However, it has been proved that two wattmeters are sufficient to measure the total three phase power , provided the wattmeters are connected as shown in Fig.6.

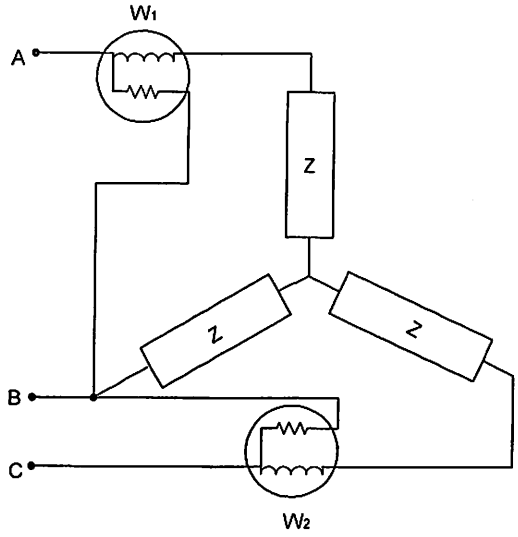


Fig. 6: The two-wattmeter method for Measuring the power in 3-phase circuits.

The total power is equal to the algebraic sum of the readings of the two wattmeters.

Due to different load power factors, one of the two meters may have negative value. In this case, its voltage or current coil connections must be reversed and the total power is equal to the difference between the two-wattmeter readings.

Procedure I: Three Phase Loading

1. Connect the circuit as shown in Fig. 7.
2. Set **Ran = Rbn = Rcn = 100Ω** by using the digital meter.

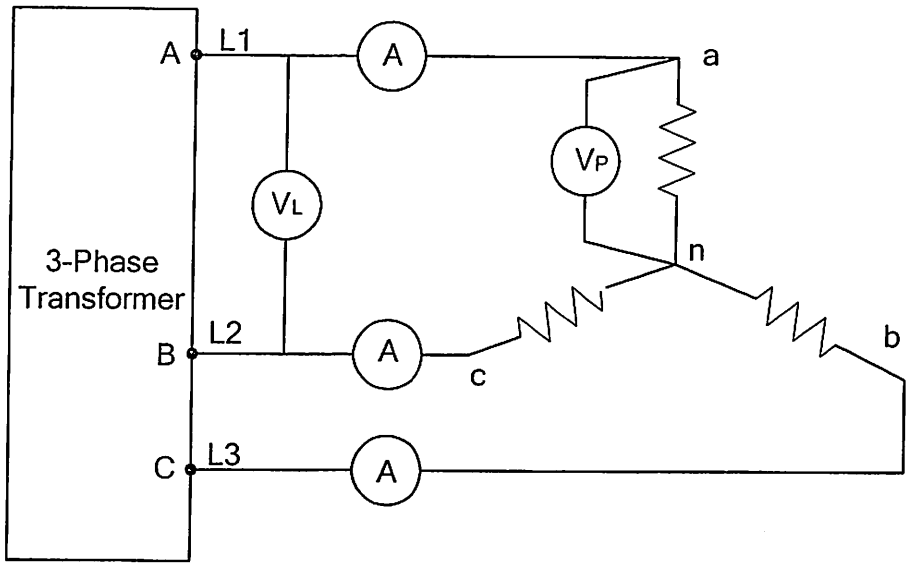


Fig. 7: Three-phase balanced load.

3. Switch on the supply.
4. Watch the readings on the three ammeters and record it.
5. Record the readings of the voltmeters.

Procedure II: Three-Phase Power Measurement Using Single Wattmeter

1. Connect the circuit as shown in Fig. 8, without changing the values of resistance which are set for a balanced load.
2. Switch on the supply.

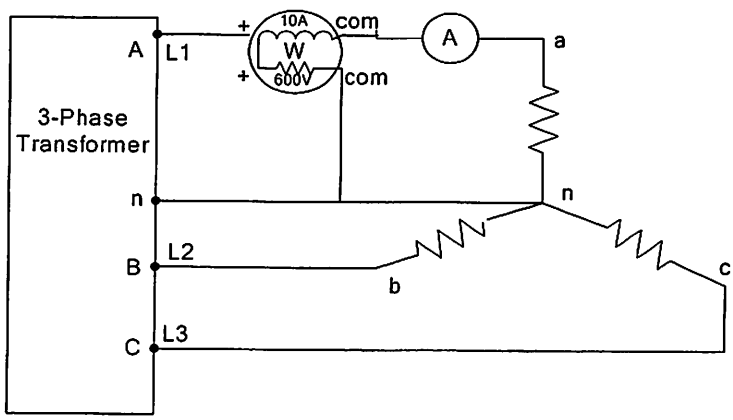


Fig. 8: Measuring Three-phase power using a single wattmeter.

3. Take readings of Power in phase A.
4. Calculate the total power by multiplying the power determined in phase A by 3.
5. Compare the total power determined above with that obtained from the expression:

$$P_T = 3\, I^2_A\, R_A$$

Procedure III: Three Phase Power Measurement Using a single Wattmeters

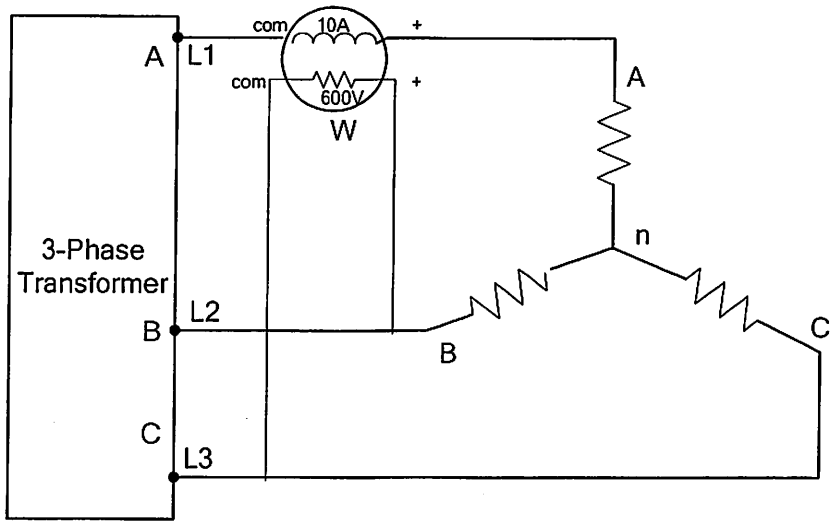


Fig. 9: Measuring the three-phase power using a single wattmeters.

1. Connect the wattmeter to measure the power in the circuit of the previous experiment as shown in Fig. 9.
2. Take readings of the three phase total power by pressing a special key on the wattmeter.
3. Compare this total power with that obtained using single wattmeter method above.

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EE-205: Electrical Circuit Lab. Experiment No. 6

Transients in DC Circuits

Objectives

To investigate the step response of a first-order electrical circuit (charging & discharging transients in RC circuits).

To investigate the step response of a second-order electrical circuit (RLC circuit) and the effect of the circuit resistance on the response.

Material Required

Resistors: A decade resistance box (10 kΩ)
Capacitors: 0.047 μF.

Inductors: One 17.5 mH.

Equipment: One audio-frequency square wave generator and a dual beam oscilloscope.

Background:

A. First Order Circuit

In an electrical system, transient is manifested when a Sudden change occurs in circuit conditions. The knowledge of the transient response of thc circuit is very important in many cases, transient

response in a first order electrical system consisting of an RC circuit will be investigated in this experiment.

i) Capacitor Charging:

When a dc voltage is switched into a series RC circuit as shown in Fig. 1 (switch S on position 1), current from the source begins to charge the capacitor.

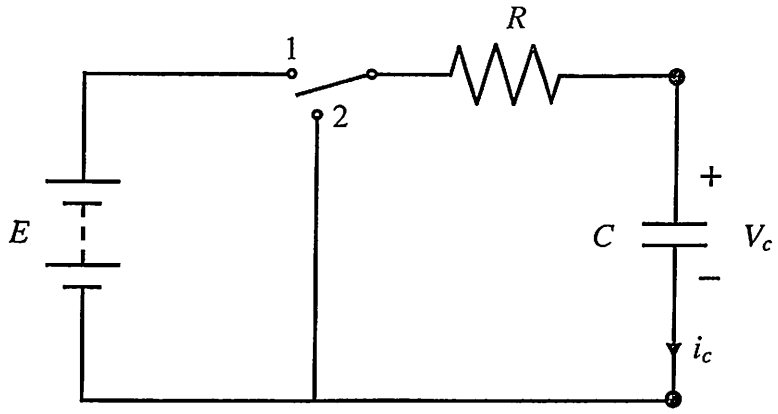


Fig.1: Transient in series RC circuit

The voltage $v_c(t)$ on the capacitor increases and the current $i_c(t)$ decreases until the capacitor C is fully charged with voltage $v_c(t)$ being equal to E and with zero current in the circuit. This state Is called the steady state. The period from switching ON the Circuit to the steady-state condition is called the transient period. During this period $v_c(t)$ and $i_c(t)$ are given as:

$$V_c(t) = E(1-e^{-t/\tau})\ V \qquad (1)$$

$$i_c(t) = (E/R)e^{-t/\tau}\ A \qquad (2)$$

where $\tau = RC$ is called the time constant of the circuit. The plot of $v_c(t)$ and $i_c(t)$ against time is shown in Figs. 2a and 2b respectively. Note that at $t = 0$, $v_c(t) = 0$ and $i_c(t) = E/R$. When $t = 5\tau$, $v_c(t) = 0.993E \approx E$ and $i_c(t) = 0.0067\ E/R = 0$. Thus a capacitor gets almost fully charged in time 5τ and transient period ends since the current almost attains steady state for $t > 5\tau$.

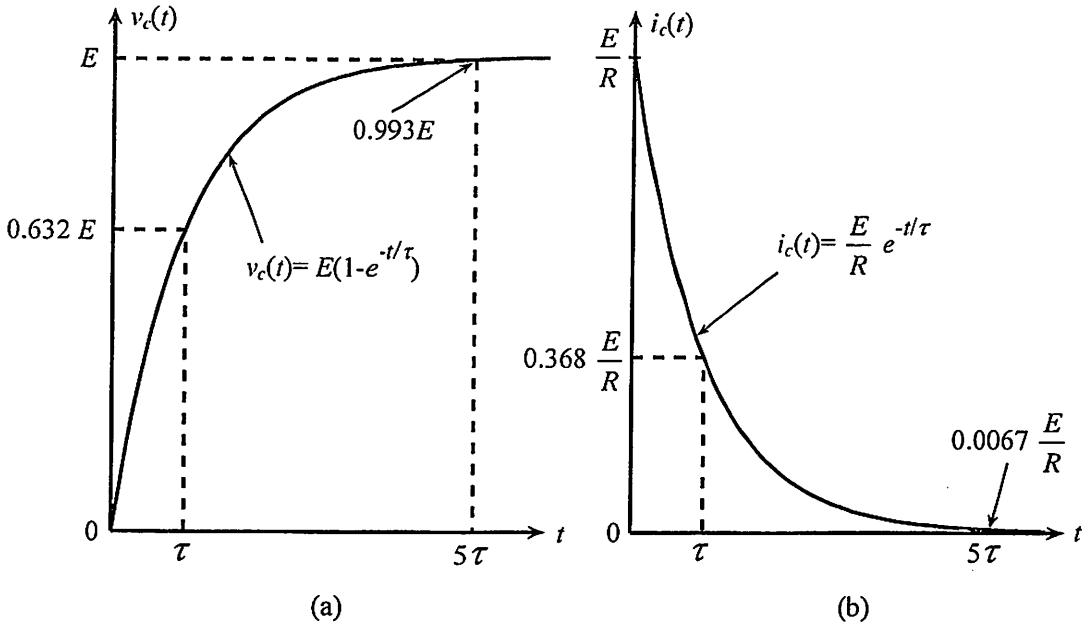


Fig.2: Plot of $v_c(t)$ and $i_c(t)$ with time.

ii) Capacitor Discharging:

After the capacitor is fully charged the switch S in Fig.1 Is thrown to position 2. In this position the capacitor starts discharging. Equations of $v_c(t)$ and $i_c(t)$ in discharging mode are written as:

$$v_c(t) = Ee^{-t/\tau} \quad (3)$$

$$i_c(t) = (-E/R)e^{-t/\tau} \quad (4)$$

The plots of voltage and current are shown in Figs. 3a and 3b. respectively.

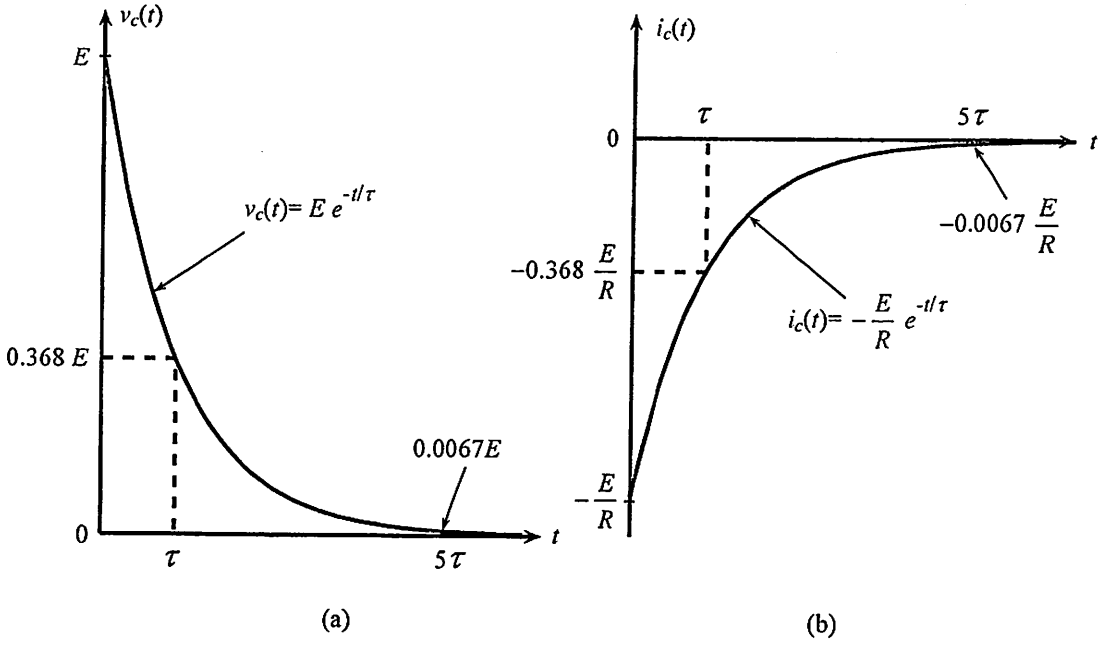


Fig.3: Plot of $v_c(t)$ and $i_c(t)$ with time.

B. Second Order Circuit

Consider a second order circuit consisting of a resistor R. an inductor I and a capacitor C, being all connected in series to a dc voltage source E as shown in Fig.4. Using KVL. one can write.

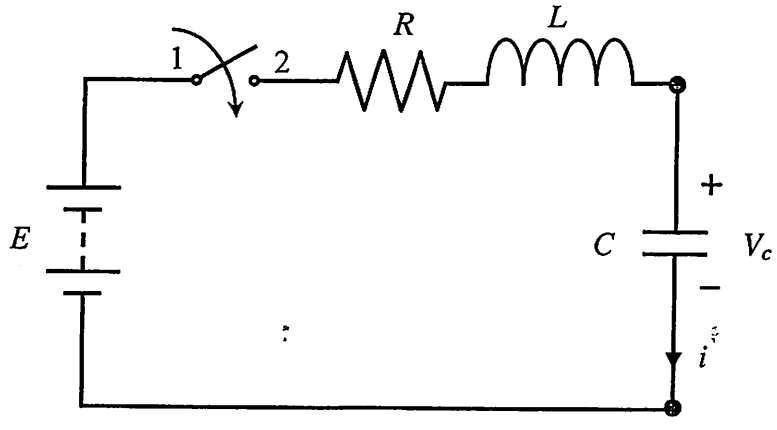


Fig.4: Series *RLC* circuit

$$Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)dt = E \qquad (5)$$

Equation (5) can be written as a second order differential equation with following roots s_1 and s_2.

$$s_1, s_2 = -\frac{R}{2L} \pm \sqrt{(\frac{R}{2L})^2 - \frac{1}{LC}} = -\alpha \pm \sqrt{\alpha^2 - \omega_o^2} \qquad (6)$$

where $\omega_o = \frac{1}{\sqrt{LC}}$, $\alpha = R / (2L)$

ω_o, in rad/s, is called the natural frequency, α, in nepers/s, is called the damping factor.

Depending upon the nature of roots s_1 and s_2, four cases may arise. These are listed below:

Case i: Lossless or Undamped Circuit

If $R = 0$, $\alpha = 0$ and the two roots s_1 and s_2 are imaginary $s_1 = j\omega_o$ and $s_2 = -j\omega_o$. Consequently v,(t) and i(t) could be written as

$$V_c(t) = E + B_1' \cos(\omega_o t) + B_2' \sin(\omega_o t) \qquad (9)$$

$$\text{i(t) = E } i(t) = B_3' \cos(\omega_o t) + B_4' \sin(\omega_o t) \qquad (10)$$

Where B_1', B_2', B_3', B_4' are constants depending on the initial conditions.

Figure 5 shows the waveforms of $v_c(t)$, and i(t) for this case. It should be noted that this case is a hypothetical case because in practice it is difficult to have zero circuit resistance.

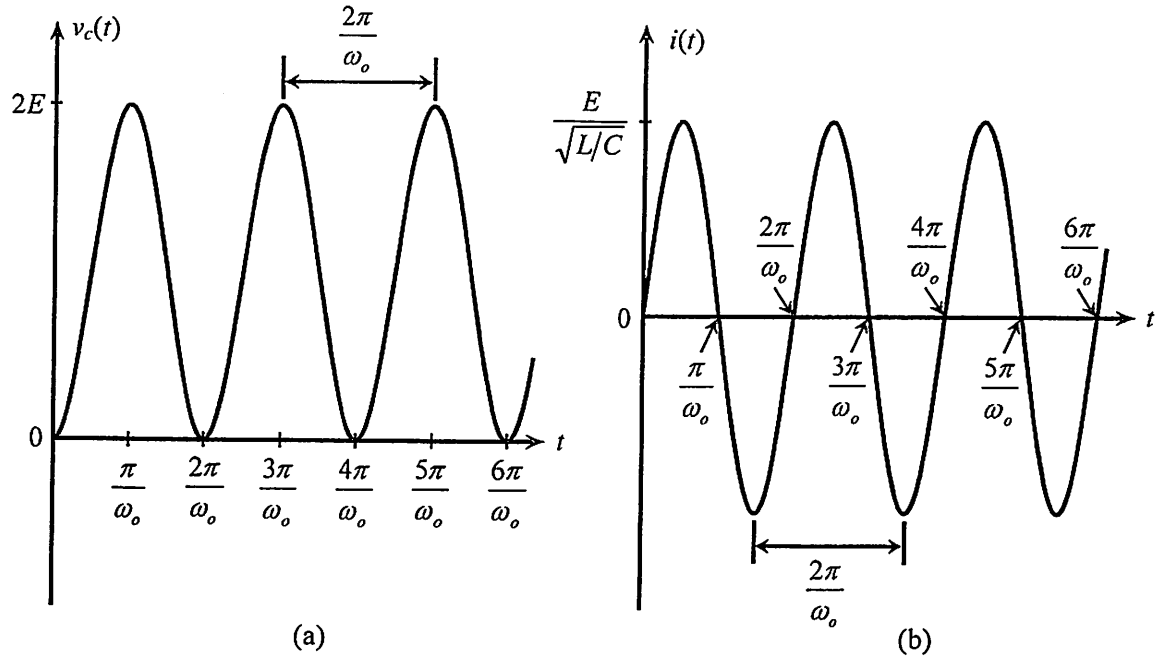


Fig.5: $v_c(t)$ and $i(t)$ for lossless or undamped case.

Case ii: Low loss or underdamped circuit

If α^2 is less than ω_o^2, (i.e. $\alpha < \omega_o$), s_1 and s_2 *are* complex and given as:

$$s_1,\ s_2 = -\alpha \pm j\omega_d \qquad (11)$$

Where $\omega_d = \sqrt{\omega_o^2 - \alpha^2}$

ω_d, in rad/s, is called the damped natural frequency.

Consequently, $v_c(t)$ and $i(t)$ could be given as

$$V_c(t) = E + e^{-\alpha t}\left[B_1 \cos(\omega_d t) + B_2 \sin(\omega_d t)\right] \qquad (12)$$

$$i(t) = e^{-\alpha t}\left[B_1' \cos(\omega_d t) + B_2' \sin(\omega_o t)\right] \qquad (13)$$

Where $B_1', B_2', B_1' and\ B_2'$ are constants depending on the initial conditions.

The response curves of voltage and current for this case are shown in Fig.6. Note that the response is very fast with an overshoot.

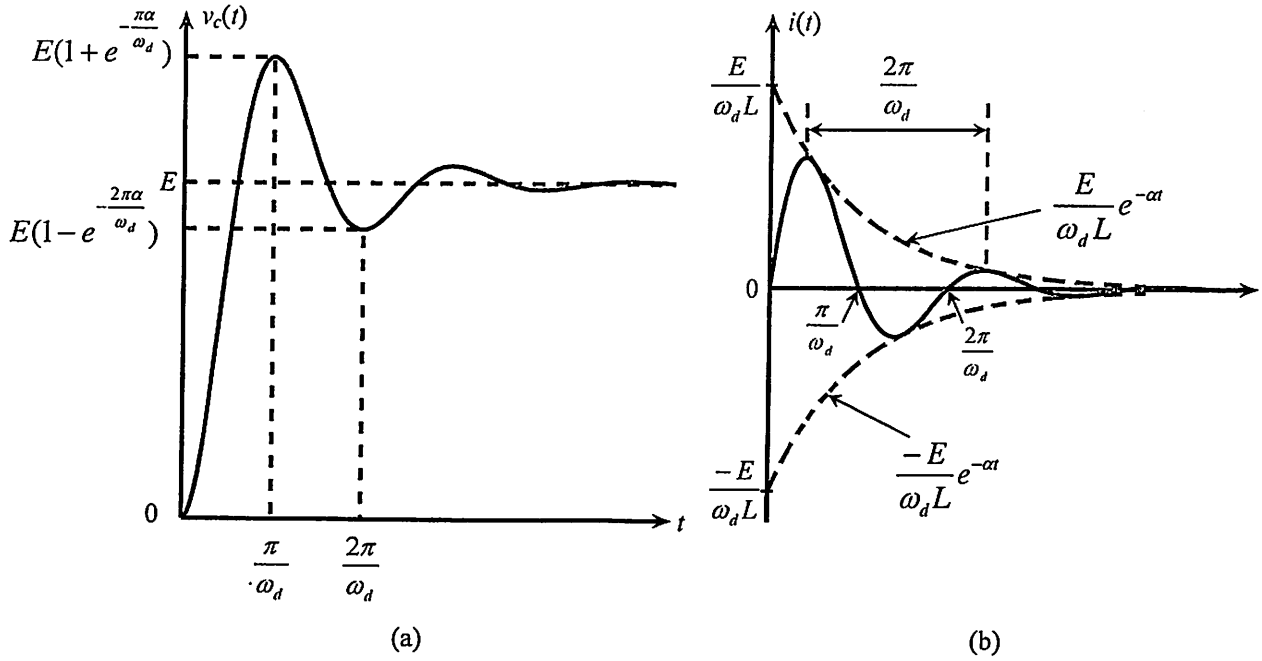


Fig.6: $v_c(t)$ and $i(t)$ for underdamped case.

Case iii: Critically Damped Circuit

1f $\alpha^2 = \omega_o^2$, ($\alpha = \omega_o$), $s_1 = s_2 = -\alpha$.

$$V_c(t) = E + (D_1 t + D_2)e^{-\alpha t} \qquad (14)$$

$$i(t) = (D_1' t + D_2')e^{-\alpha t} \qquad (15)$$

Where $D_1, D_2, D_1' and D_2'$ are constants depending on the initial conditions.

The response curves of voltage and current are shown in Fig.7. Note that the response is fast

without overshoot. The resistance for which critical damping occurs is called critical resistance and is defined from:

$$\frac{R}{2L}=\frac{1}{\sqrt{LC}}\Rightarrow R^2LC=4L^2\Rightarrow R=\sqrt{\frac{L}{C}}$$

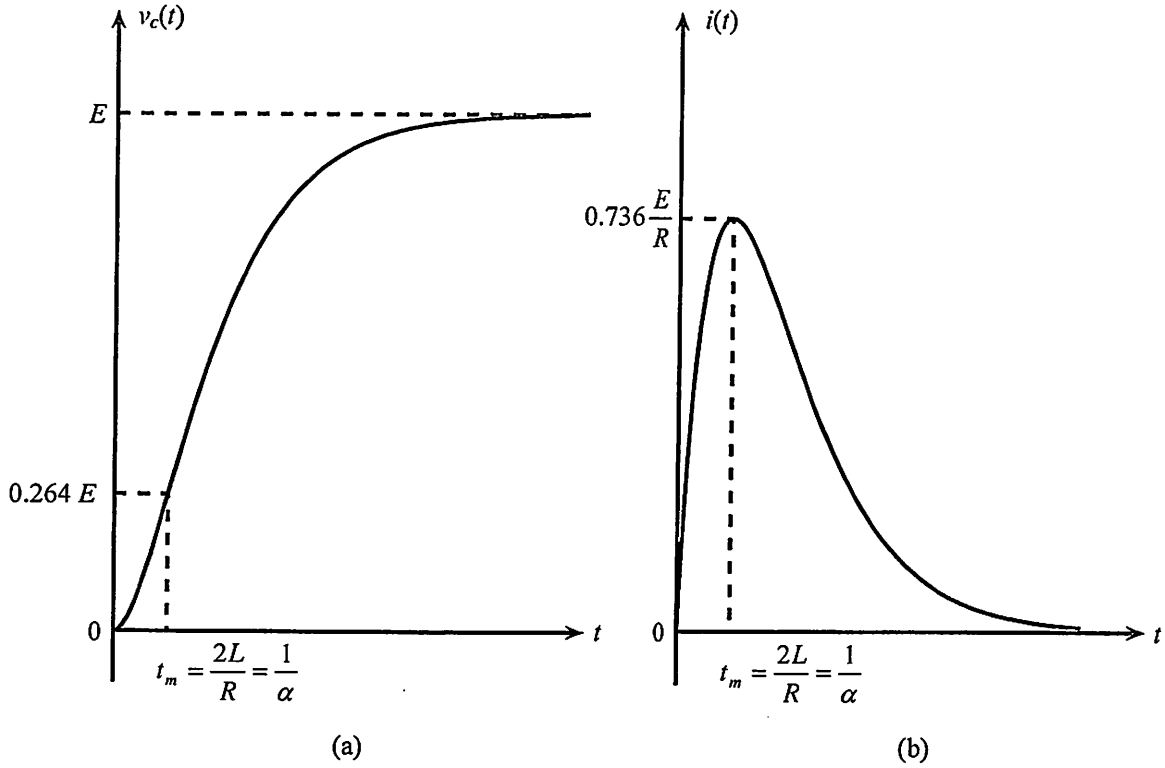


Fig.7: $v_c(t)$ and $i(t)$ for critically damped case.

Case iv: Over Damped or High-Loss Circuit

If $\alpha^2 > \omega_o^2$, (i.e. $\alpha > \omega_o$), s_1 and s_2 are real and unequal and given as:

$$s_1,\ s_2 = -\alpha + \beta$$

$$\beta = \ =\sqrt{\alpha^2-\omega_o^2}$$

Consequently, $v_c(t)$ and $i(t)$ can be written as

$$V_c(t) = E + A_1 e^{s_1 t} + A_2 e^{s_2 t} \qquad (16)$$

$$i(t) = A_1' e^{s_1 t} + A_2' e^{s_2 t} \qquad (17)$$

Where $A_1, A_2, A_1' and A_2'$ are constants depending on the initial conditions.

The response curves of voltage and current are shown in Fig.8. Note that the response is slow as compared to the critical damped case.

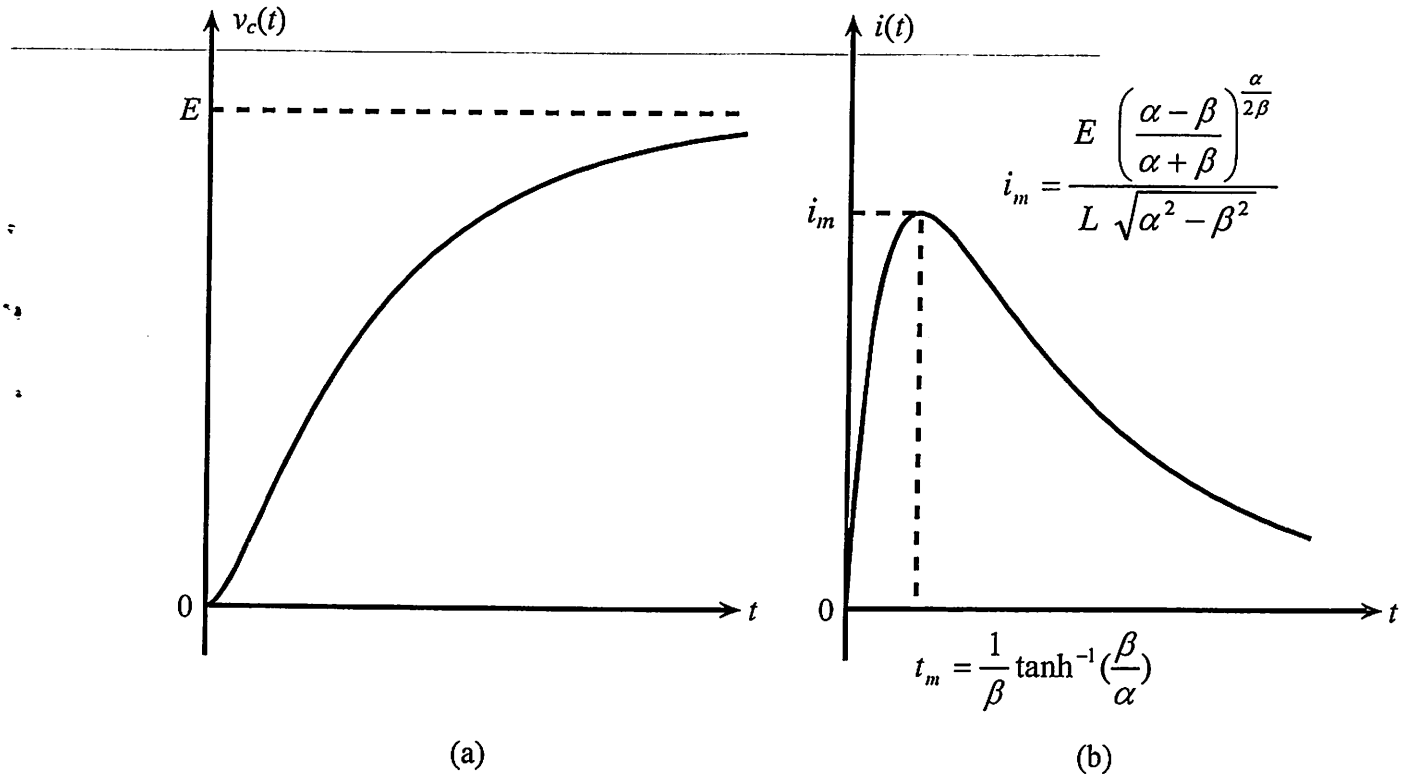


Fig.8: $v_c(t)$ and $i(t)$ for overdamped case.

Procedure I:

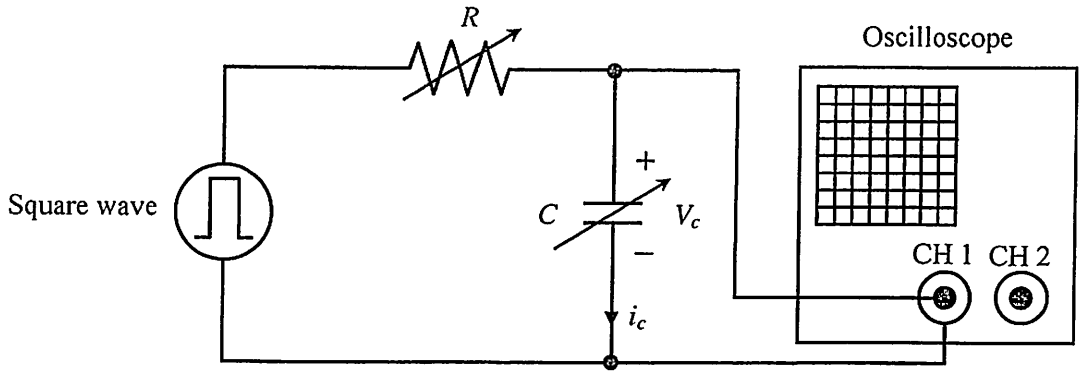


Fig. 9:Expermental setup.

1. Connect the circuit as shown in Fig.9 and set the square wave voltage that alternates between 0 V and +10 V with a frequency of about 200 Hz. Set the decade resistance box to 10 kΩ. Note that R_s is the internal resistance of the square wave generator which is to be added to R for computation.

2. Display the capacitor voltage on the oscilloscope. Select a suitable time base to get full view of the waveform (rising edge and steady-state portion). Adjust the frequency of square wave oscillator, if necessary. Record $v_c(t)$ for six different times in Table 1. Also measure the time constant and write its value in Table 2.

3. Display the discharging waveform of the capacitor voltage on the oscilloscope and adjust its frequency such that full decaying portion of $v_c(t)$ appears on the screen. Measure t and $v_c(t)$ at six different times in the decay portion and record your results in Table 1. Also measure the time constant and write its value in Table 2.

4. For sets (2) and (3) compare the theoretically calculated values of $v_c(t)$ and τ with measured values for different cases and complete Tables 1 and 2.

5. Using $v_c(t)$ waveform obtained in steps (1) and (3) for charging and discharging modes, find waveform of $v_R(t)$ and $i_c(t)$ corresponding to R = 10 kΩ

Table 1: Experimental and theoretical values of $v_c(t)$

Circuit Element	Time	Charging Mode			Time	Discharging Mode		
		$v_c(t)$				$v_c(t)$		
		Experimental	Theoretical	% Difference		Experimental	Theoretical	% Difference
R = 10 kΩ C = 0.047 μF								

Table 2: Time constants of the RC circuits.

Circuit Element	Charging Mode			Discharging Mode		
	Time Constant		error %	Time Constant		error %
	Experimental	Theoretical		Experimental	Theoretical	
R = 10 kΩ C = 0.047 μF						

Procedure II

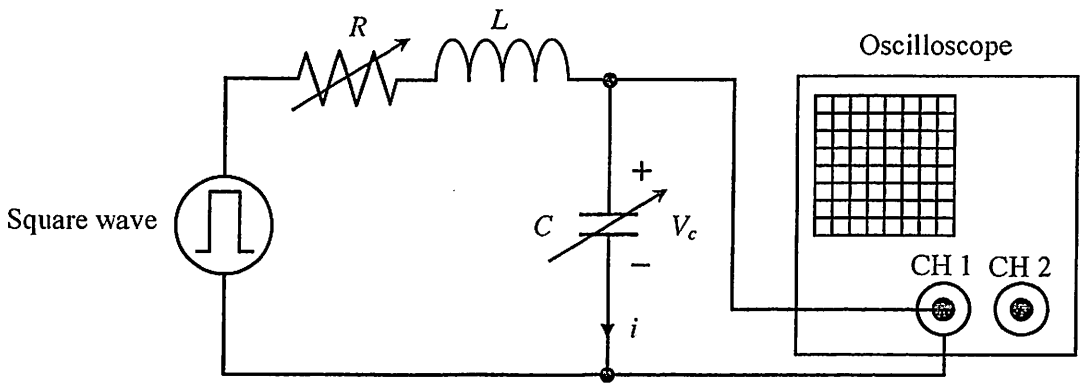


Fig.10: Exprimental setup.

1. Connect the circuit as shown in Fig.10 and set the square wave voltage that alternates between 0 V + 10 V with a frequency of about 200 Hz and set the decade resistance box to suitable value, say, 600 Ω, while Capacitance is 0.047 μF, respectively, to give under damped response. Rs is the internal resistance of the square wave generator which should be added to R for computation.

3. Display the voltage across the capacitor on the oscilloscope. Select a suitable time base to get the full view of the waveform (rising edge and steady state portion). Adjust the frequency of square wave oscillator, if necessary. Increase the value of R till the overshoot just vanishes. The value of resistance at which the overshoot just vanishes is the critical resistance ($R_c = 2\sqrt{L/C}$). Increase the resistance further to have over damped response. Trace the above three curves on the same sheet and record the observations.

King Saud University

College of Engineering

Electrical Engineering Department

EE-205: Electrical Circuit Lab Experiment No.7

Magnetically Coupled Circuits

Objectives

To get acquainted with the basic principles of magnetic coupling, self inductance, mutual inductance, coupling coefficient.

Required Equipment

Coils: Two coupled coils (500 and 1000 tuns) connected as a transformer.

Equipment: Two Ammeters, three Voltmeters, Variable AC Power Supply (Variac).

Background

I. Self Inductance

Consider a single coil wound around a magnetic core as shown in Fig. 1.

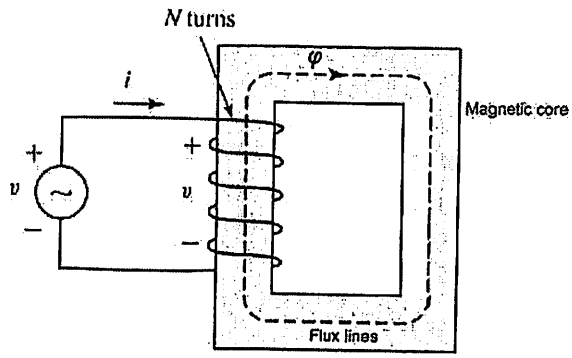


Fig. 1: A single coil wound around a magnetic core.

A current i flowing in this coil will produce a flux ϕ in the magnetic core. If the current is varying with time, the produced flux will also be varying with time. According to Faraday's law of induction, an induced voltage v will be generated across the coil terminals. V is given by the relation,

$$v = N \frac{d\phi}{dt} \quad (1)$$

The self inductance of a coil is defined as the ratio between the flux linkage, $N\phi$, linking the coil and the current, i , producing the flux, or:

$$L = \frac{N\Phi}{i} \quad (2)$$

Substitution from equation (2) into (1) gives

$$v = L\frac{di}{dt} \quad (3)$$

In this equation L is assumed constant, otherwise the following general equation has to be used.

$$v = \frac{d}{dt}(Li) \quad (4)$$

II. Mutual Inductance

If a second coil is wound around the magnetic core of Fig.1 as shown in Fig.2, the two coils become magnetically coupled. The flux produced by the first links the second. The mutual inductance, M_{12}, between coil 1 and coil 2 is defined as the ratio between the flux linkage,

$N_2\phi$, linking coil 2 and the current i_1 in coil 1 producing this flux, or:

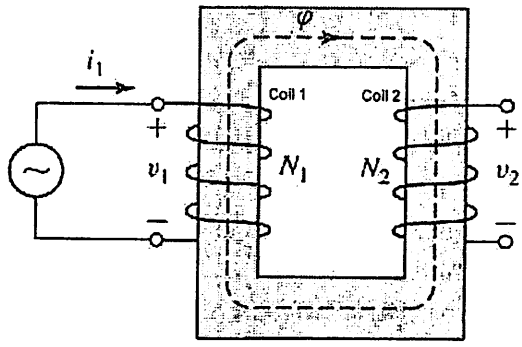


Fig. 2: Two magnetically-coupled coils.

$$M_{12} = \frac{N_2 \Phi}{i_1} \tag{5}$$

When the terminals of coil 2 are open-circuited, i.e. no current flows in this coil, the following equations can be written.

$$v_1 = \frac{d}{dt}(N_1 \Phi) = L_1 \frac{di_1}{dt} \tag{6}$$

$$v_2 = \pm \frac{d}{dt}(N_2 \Phi) = \pm M_{12} \frac{di_1}{dt} \tag{7}$$

v_2 is the voltage induced in coil 2 due to the current i_1 in coil 1. It can be shown that $M_{12} = M_{21}$, subscripts are unimportant and we can write:

$$M_{12} = M_{21} = M \tag{8}$$

The polarity of the induced voltage v_2 depends on the relative method of coiling. To explain this point look at the 3 coils of Fig. 3, which are wound on the same magnetic core.

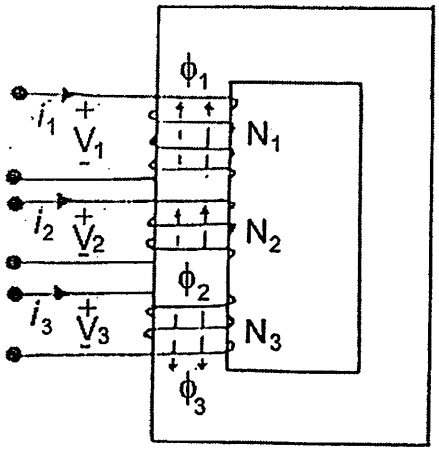
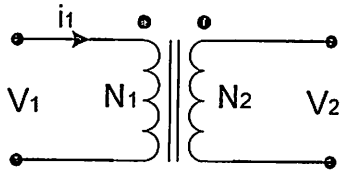


Fig. 3: Three magnetically-coupled coils.

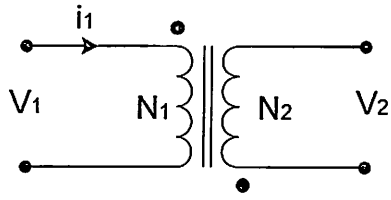
From this figure, it can be seen that coil 3 is wound differently. It follows that v_1 and v_2 have the same polarity, while v_3 has a different polarity. Notice also that a current flowing in coil 1 in the direction shown will set a magnetic field in the same direction as that produced by coil 2. The magnetic field produced by coil 3 when a current flows into it in the direction shown will be opposing that of coils 1 or 2.

III. The Dot Rule

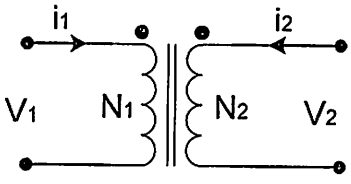
The polarity of the induced voltage is very important when writing the equations of an electric circuit. To indicate this, a dot is put at one of the terminals of each coil to show its relative polarity as shown in Fig.4.



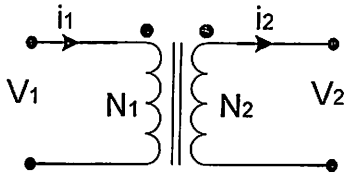
(a) V_2 has the same polarity as V_1.



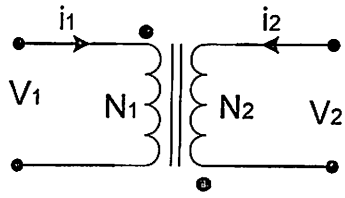
(b) V_2 has an opposite polarity to that of V_1.



(c) mmf of coil 2 adds to that of coil 1.



(d) mmf of coil 2 opposes that of coil 1.



(e) mmf of coil 2 opposes that of coil 1.

Fig. 4: Application of dot rule.

IV. Mathematical Relations

The mathematical relations describing mutually coupled circuits depend on the method of connection.

For series connected coils, two situations are met. The first is the addition situation while the second is the opposition situation . Both cases are shown in Fig.5.

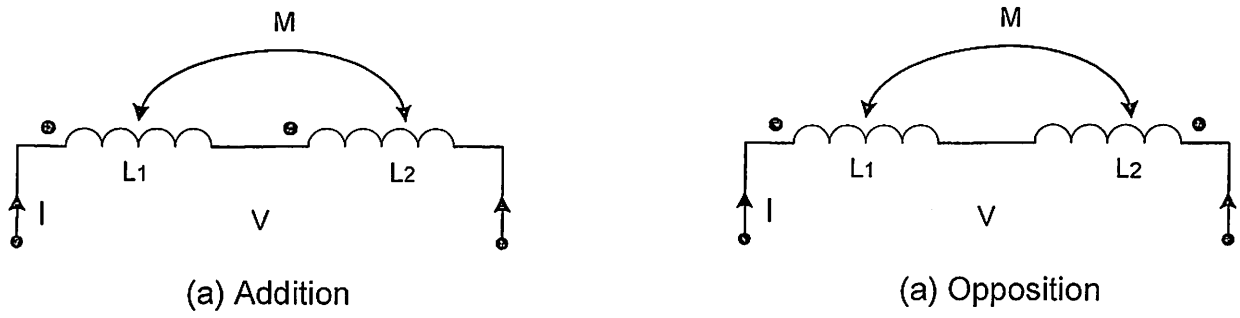


Fig. 5: Two magnetically-coupled coils connected in series.

The corresponding voltage equation for each case, when coil resistance are neglected is

$$v = (L_1 + L_2 + 2M)\frac{di}{dt} \qquad \text{(Additive)} \qquad (9)$$

$$v = (L_1 + L_2 - 2M)\frac{di}{dt} \qquad \text{(Opposition)} \qquad (10)$$

For parallel connected circuits as that shown in Fig.6, the following relations can be written:

$$v = L_1\frac{di_1}{dt} + M\frac{di_2}{dt} \qquad (11)$$

$$v = L_2\frac{di_2}{dt} + M\frac{di_1}{dt} \qquad (12)$$

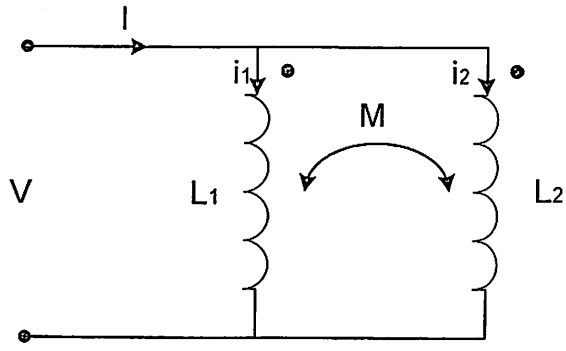


Fig. 6: Parallel connected magnetically coupled circuit.

It can be easily proved that the equivalent inductance of this circuit is given by:

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \qquad (13)$$

V. Coefficient of Coupling

Not all the flux linking a coil flows inside the magnetic core. A small percent of the flux produced by a coil leaks away from the core. This is known as the leakage flux. It follows that the self inductance of a coil has a value higher than that of the mutual inductance between it and an identical coil. The measure of the leakage flux is the coupling coefficient k.

$$k = \frac{M}{\sqrt{L_1 L_2}} \qquad (14)$$

The maximum value of the coupling coefficient is 1 for the ideal case of no leakage. In this case:

$$M = \sqrt{L_1 L_2} \qquad (15)$$

VI. AC Excitation

For sinusoidal applied voltage of the form $V = V_m \cos \omega t$ the steady state currents will be also sinusoidal. To take care of this, the operator d/dt in equations (9-12) is replaced by $j\omega$ and the following relation can be written:

$$\overline{V} = [(r_1 + r_2) + j\omega\{L_1 + L_2 + 2M\}]\overline{I} \qquad (16)$$

for series connection (addition) and

$$\overline{V} = [(r_1 + r_2) + j\omega\{L_1 + L_2 - 2M\}]\overline{I} \qquad (17)$$

for series connection (subtraction). V and I are, respectively, the phasor voltage and the phasor current.

Notice that we have considered here the coil resistance; r_1 for coil 1 and r_2 for coil 2.

For a single coil, following equation applies:

$$\overline{V} = (r + j\omega L)\overline{I} = \overline{ZI} \qquad (18)$$

where $\overline{Z}$ = r + jωL is the impedance of the coil at the frequency ω rad/s.

Procedure I - Determination of Self Inductance

1. Measure the resistance R_{dc} of coil 1 with the help of multimeter. This is dc resistance of coil 1
2. Determine the coil 1 ac resistance R_1 by allowing 15% increase due to skin effect, such that

$$R_1 = 1.15\ R_{dc}$$

3. Similarly find out R_2, the ac resistance of coil 2.
4. Connect coil 1 to a variable 60 Hz voltage source as shown in Fig. 7.

Coil 1: 1000 turns.

Coil 2: 500 turns.

5. Adjust the magnitude of the applied voltage to 2 Vrms and then take the readings of both the ammeter and voltmeter.
6. Calculate the coil (self) inductance as follows:

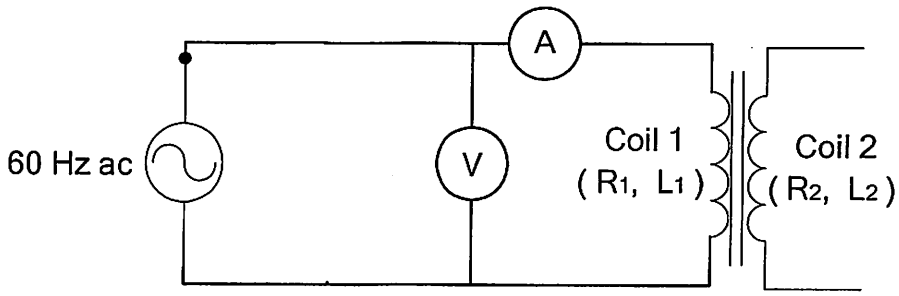


Fig. 7: Measuring the self inductance of a coil.

$$Z_1 = \frac{V}{I} = \sqrt{R_1^2 + X_1^2}$$

$$X_1 = \sqrt{Z_1^2 - R_1^2}$$

$$L_1 = \frac{X_1}{\omega}$$

7. Repeat for coil 2 to find L_2.

Procedure II - Determination of Mutual Inductance

1. Connect the circuit as shown in Fig. 8, with coil 2 magnetically coupled to coil 1.

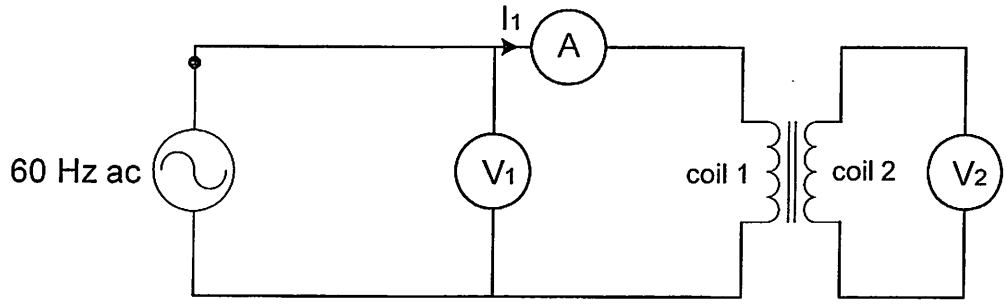


Fig. 8: Determination of the mutual inductance.

2. Adjust the magnitude of the applied voltage to 2 Vrms, and take the readings of V_1, I_1 and V_2.

3. Calculate the mutual inductance as follows:

$$X_M = V_2 / I_1$$

$$M = X_M / \omega$$

Procedure III - Determination of Polarity of Magnetically Coupled Coils

1. Connect the electric circuit as shown in Fig. 9.

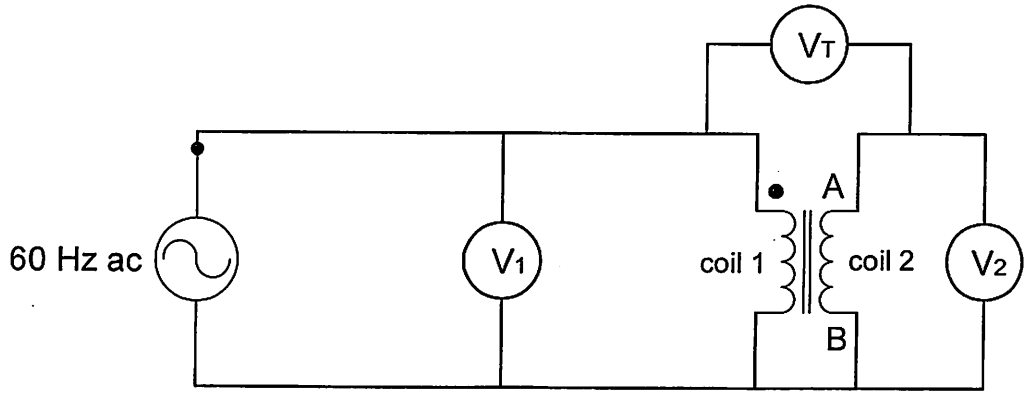


Fig. 9: Determination of the polarity.

2. With specified voltage applied to coil 1, take the reading of V_T.

3. If V_T is the difference between V_1 and V_2, the dot is to be put at point A, otherwise it is to be put at point B.

Procedure IV - Series Connected Magnetically Coupled Coils

1. According to the polarity determined above, connect the electric circuit as shown in Fig.10.

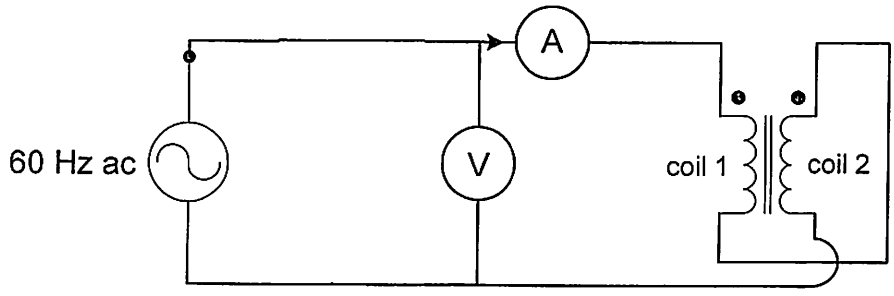


Fig. 10: Series-connected magnetically -coupled coils (adding).

2. Adjust the supply voltage to 2 Vrms and take the readings of the voltage and current.

3. Calculate the equivalent impedance in this case as follows:

$$X_{eq1} = \sqrt{(V/I)^2 - (R_1 + R_2)^2}$$

$$= \omega (L_1 + L_2 + 2M)$$

4. Repeat steps 1 and 2 but with the two coils connected in opposition

$$X_{eq2} = \sqrt{(V/I)^2 - (R_1 + R_2)^2}$$

$$= \omega (L_1 + L_2 - 2M)$$

5. Determine the mutual inductance M_{12} using the relation:

$$M_{12} = \frac{X_{eq1} - X_{eq2}}{4\omega}$$

6. Check with the value obtained before and comment on any discrepancy between these results.

7. Determine the coupling coefficient k. Where,

$$k = \frac{M_{12}}{\sqrt{L_1 L_2}}$$

King Saud University,

College of Engineering,

Electrical Engineering Department

EE – 205: Electrical Circuit Lab. Experiment No. 8

Resonance in Series and Parallel Circuits

Objectives

- Investigation of the frequency response of series RLC and parallel RLC circuits.
- Familiarization with the concepts of resonance, quality factor, and bandwidth

Required Equipment

Resistors: 100Ω, 10Ω, 5 k Ω (Two).

Inductors: 17.5 mH (1000 turns air core), 4.38mH (500 turns/air core).

Capacitors: 0.1 μF, 4.7 μF, 1 μF, 10 nF

Equipment: Two multimeters, CRO, and function generator

Background

I. Series RLC Circuit

The impedance of the series RI.C circuit ot Fig.I-a is:

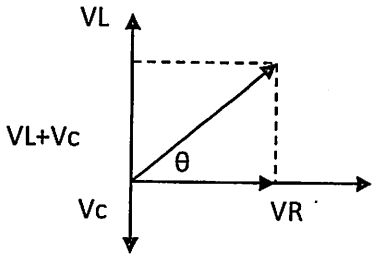


Fig.1:a

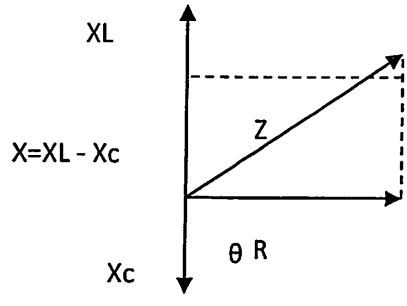


Fig.1:b

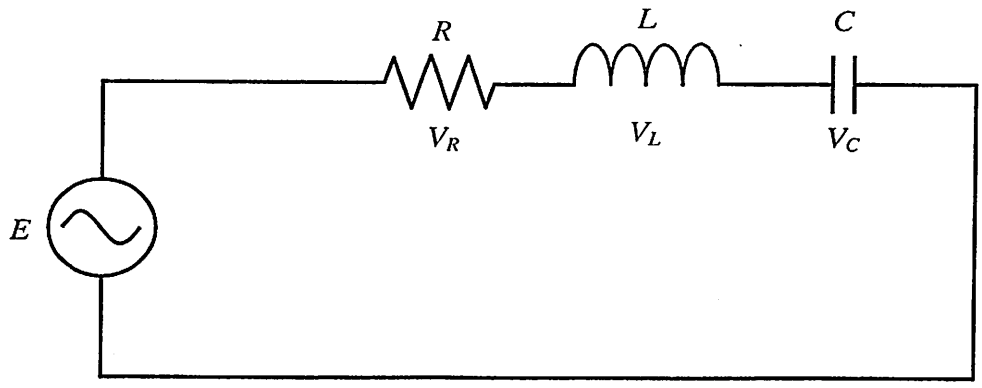


Fig.1:C

Z = magnitude of the *impedance* = $\sqrt{R^2 + X^2} = \sqrt{R^2 + (X_L - X_C)^2}$ (1-b)

θ = angle *of the impedance* = $\tan^{-1} X / R = \tan^{-1}(X_L - X_C)/R$ (1-c)

$X_L = \omega L$ = inductive reactance i.e. reactance of L.

$X_C = 1/\omega C$ = capacitive reactance i.e. reactance of C.

From Fig.3-b, the phasor voltages are related by:

$$\overline{E} = \overline{V}_R + \overline{V}_X = \overline{V}_R + \overline{V}_L + \overline{V}_C \quad \text{(2-b)}$$

$$E = \sqrt{V_R^2 + (V_L - V_C)^2} \quad \text{(2-c)}$$

where

V_R = magnitude of the voltage across R

V_L = magnitude of the voltage across L

V_C = magnitude of the voltage across C

$\overline{I} = \overline{E}/\overline{Z}$ = phasor current (3)

$F_P = \cos\theta = P/EI = R/Z$ (4)

where

F_P = power factor

P = average (active) power = $EI\cos\theta = I^2R$

θ = the angle of Z

E = magnitude of the total voltage

I = magnitude of the current

EI = apparent power

$EI\sin\theta$ = reactive power = $I^2X = I^2X_L + I^2X_C$

The impedance Z can be varied (magnitude and phase) by varying one of the elements (R, L, and C) and/or varying the frequency. In many practical situations this is achieved by, varying the frequency.

$$\overline{Z} = R + j(\omega L - \frac{1}{\omega C}) \qquad \text{(5-a)}$$

$$Z = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2} \quad , \theta = \tan^{-1}\frac{\omega L - 1/\omega C}{R} \qquad \text{(5-b)}$$

Figure 2 shows the variation of the impedance components [resistive component, reactive component, magnitude and phase] with the frequency. The series resonant frequency f_{os}, is defined as the frequency at which the impedance is pure resistive which requires zero reactance (X = 0) or unity power factor.

$$\omega_{os} = 1/\sqrt{LC} \quad \text{rad/sec} \quad \text{or} \quad f_{os} = 1/2\pi\sqrt{LC} \quad Hz \qquad \text{(6-a)}$$

The impedance has a capacitive reactance for frequencies lower than the resonant frequency foe and an inductive reactance for frequencies higher than f_{os}. The magnitude of the impedance Z is minimum and equal to R at f_{os} and increases for frequencies lower or higher than f_{os},. The two frequencies f_1 and f_2 at which Z is $\sqrt{2}$ R is of particular interest. At f_1,

X = -R and at f_2, X = R. It can be easily shown that:

$$f_1 = -\frac{R}{4\pi L} + \sqrt{f_{os}^2 + (\frac{R}{4\pi L})^2} \quad ; \quad f_1 < f_{os} = \frac{1}{2\pi\sqrt{LC}} \qquad \text{(6-b)}$$

$$f_2 = \frac{R}{4\pi L} + \sqrt{f_{os}^2 + \left(\frac{R}{4\pi L}\right)^2} \quad ; \quad f_2 > f_{os} = \frac{1}{2\pi\sqrt{LC}} \qquad (6\text{-c})$$

f_1, and f_2 are called the half-power frequencies or the 3-dB frequencies. The difference between f_2 and f_1, is called the frequency bandwidth β. Thus

$$\beta = f_2 - f_1 = \frac{R}{2\pi L} \quad Hz \qquad (7)$$

It is worthy to note that $f_2 f_1 = f_{os}^2$ while $(1/2)(f_2 + f_1) = f_{os}$, which means that the resonant frequency f_{os}, is the geometric mean of the half-power frequencies f_2 and f_1 but not exactly their arithmetic mean. However, if f_{os}, > > $R/4\pi L$, then f_{os}, will be almost at the center of the band width.

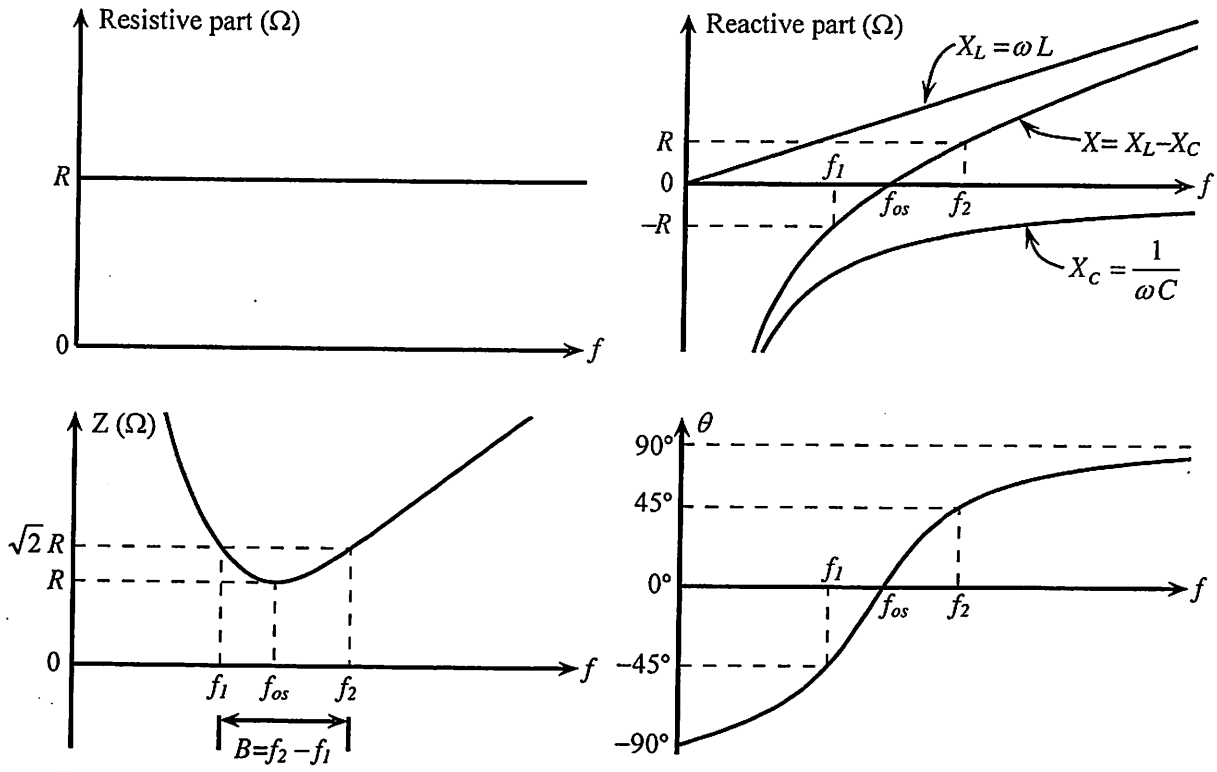


Fig.2: Impedance components of a series RLC circuit.

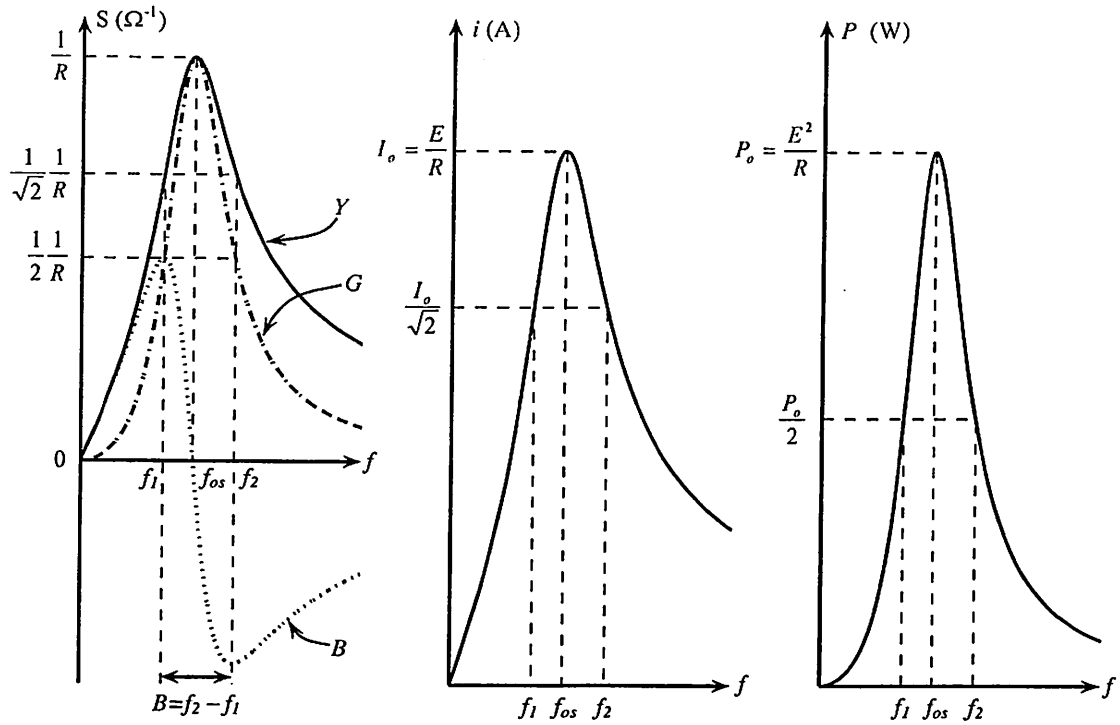


Fig.3: Admittance components, current, and average power variation with frequency in a series RLC circuit.

The admittance Y of the series RLC circuit is:

$$\overline{Y} = G + jB = \frac{1}{\overline{Z}} = \frac{1}{R + jX} = \frac{R}{R^2 + X^2} + j\frac{-X}{R^2 + X^2} \qquad \text{(8-a)}$$

Thus

$$Y = \frac{1}{\sqrt{R^2 + X^2}}, \quad \theta_Y = -\theta = -\tan^{-1} X/R, \quad G = R/(R^2 + X^2) \text{ and } B = -X/(R^2 + X^2) \qquad \text{(8-b)}$$

Figure 3-a shows the variation of the admittance components with frequency. If the series RLC circuit is driven by a constant voltage source, E = const., although of variable frequency, then the current magnitude, I = EY, is proportional to the admittance magnitude, Y. Thus the current magnitude variation with frequency is similar to the variation of Y with frequency as shown in Fig. 3-a. Also the average power dissipated is $P = EI\cos\theta = E^2 Y\cos\theta = E^2 G$ which means that P is proportional to G. Figures 3-a and b show, respectively, the variation of I and P with the frequency under constant E. It is clear that I and P are maximum at resonance while I reduces to $I/\sqrt{2}$ of its

resonant value at f_1 and f_2 while P decreases to half its resonant value at f_1 and f_2 and hence the name half-power frequencies. If we fix L and C and increase R the curves of I and P get sharper: the peak, E/R or E^2/R, gets higher and the width ($B = f_2 - f_1$) gets narrower. In such case the frequencies near f_{os} pass easily while the far frequencies, on both sides of f_{os} are highly attenuated. To measure such frequency selectivity, a quality factor Q_0 is defined as:

$Q_0 = f_{os} / B = f_{os} / (f_2 - f_1)$ = resonant frequency / frequency bandwidth (9-a)

Substituting for $B = R/2\pi L$ gives

$$Q_0 = (2\pi f_{os} L) / R = (\omega_{os} L) / R \qquad (9\text{-b})$$

Again substituting for $f_{os} = 1/2\pi \sqrt{LC}$ gives

$$Q_0 = \frac{\sqrt{L/C}}{R} \qquad (9\text{-c})$$

which shows that Q_0 is inversely proportional to the resistance R. Also, by multiplying (9-b) by I_0/I_0, where I_0 is the current at resonance, ONE GETS:

$$Q_0 = V_{L0} / E = V_{C0} / E \qquad (9\text{-d})$$

where V_{L0} and V_{C0}, which are equal, are calculated at resonant frequency ω_0. In most practical cases $Q_0 >> 1$ and thus the magnitude of the voltage across L or C at resonance may be much higher than the applied voltage E. This must be taken into account when dealing with ac circuits especially near resonance.

II Parallel RLC circuit

Figure 4 shows a parallel GCL circuit. Following the same procedure as in series RLC circuit or using the concept of duality we get:

$$\overline{Y} = G + jB \;;\; B = B_C + B_L;\; B_C = \omega C, \text{ and } B_L = -1/\omega L \qquad (10\text{-a})$$

$$Y = \sqrt{G^2 + B^2} = \sqrt{G^2 + (B_C + B_L)^2} \;;\; \theta_y = \tan^{-1} B/G = \tan^{-1} (B_C + B_L)/G \qquad (10\text{-b})$$

$$\bar{I} = \bar{I}_G + \bar{I}_B = \bar{I}_G + \bar{I}_C + \bar{I}_L \qquad (11\text{-a})$$

$$I = \sqrt{I_G^2 + (I_C - I_L)^2} \qquad (11\text{-b})$$

Where I_G, I_C and I_L are, respectively, the magnitudes of $\bar{I}_G, \bar{I}_C$ and $\bar{I}_L$

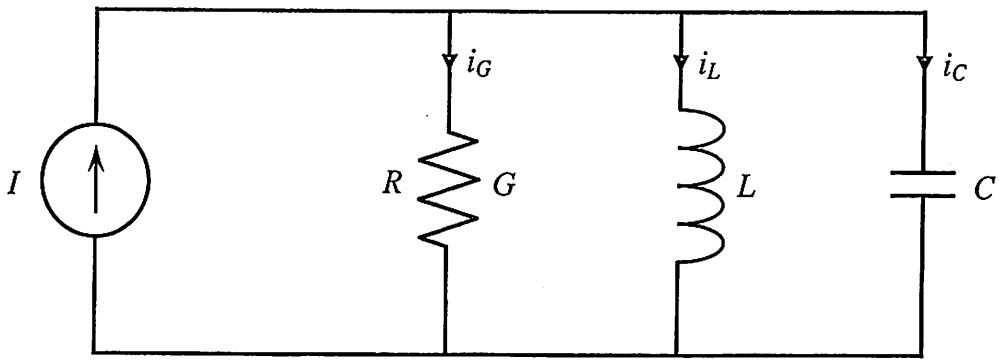


Fig.4: Parallel GCL Circuit

The variations of the admittance components with frequency match those of the impedance components in Fig.2: R→G, X → B, Z →Y and θ → - θ_Y. The parallel resonant frequency f_{op} is the frequency at which Y is pure conductive i.e. B = 0. This gives:

$$\omega_{OP} = 1/\sqrt{CL} \text{ rad/s} \quad \text{or} \quad f_{op} = 1/2\pi\sqrt{CL} \text{ Hz} \qquad (13\text{-a})$$

The half-power frequencies are

$$f_1 = -\frac{G}{4\pi C} + \sqrt{f_{op}^2 + (G/4\pi C)^2} \quad \text{Hz} \qquad (13\text{-b})$$

$$f_1 = \frac{G}{4\pi C} + \sqrt{f_{op}^2 + (G/4\pi C)^2} \quad \text{Hz} \qquad (13\text{-c})$$

The frequency bandwidth B is:

$$B = f_2 - f_1 = G/2\pi C \quad \text{Hz} \qquad (14)$$

The impedance components of the parallel GCL circuit are

$$\overline{Z} = R+jX = 1/\overline{Y} = 1/(G+jB) = G/(G^2+B^2) + J\,(-B/(G^2+B^2)) \qquad (15\text{-a})$$

$$Z = 1/\sqrt{G^2+B^2},\ R = G/(G^2+B^2),\ X = -B/(G^2+B^2) \qquad (15\text{-b})$$

If the parallel GCL circuit is driven by a constant current source I = const., although of variable frequency, then

$$V = I\,Z \propto Z \qquad (16\text{-a})$$

$$P = IV\cos\theta_Y = I^2 Z\cos\theta_Y = I^2 R \propto R \qquad (16\text{-b})$$

Figure 5 shows the variation of both V and P with frequency.

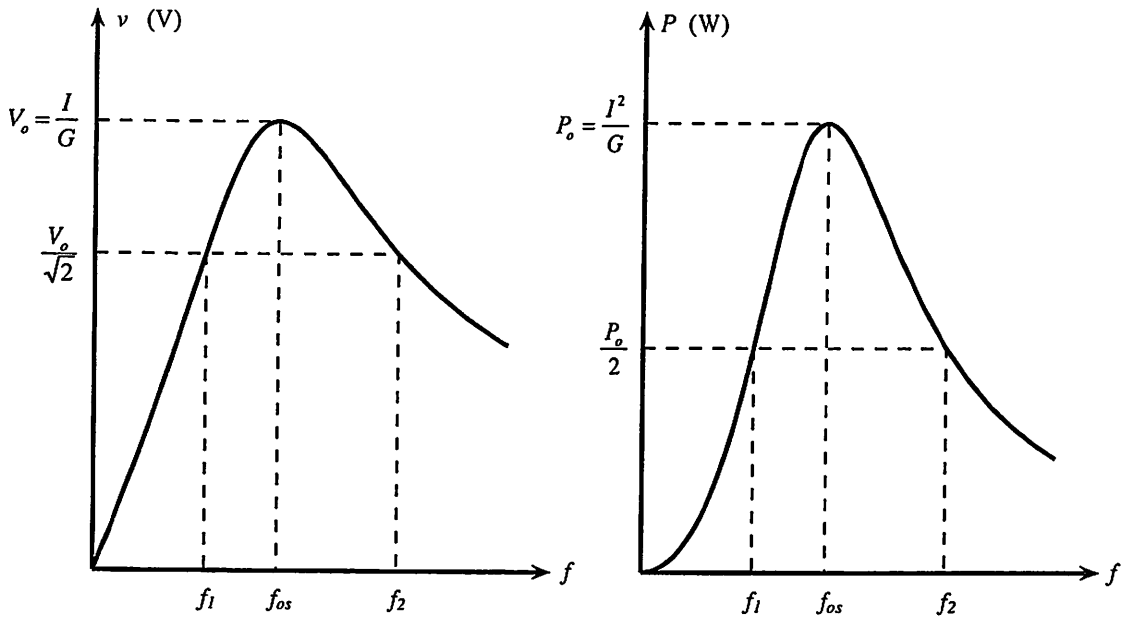


Fig.3: Voltage and average power variation with frequency in GCL circuit.

The quality factor is:

$$Q_0 = \frac{f_{op}}{B} = \frac{f_{op}}{f_2 - f_1} = \frac{\omega_0}{G} = (\sqrt{C/L})/G = \frac{I_{co}\ or\ I_{LO}}{I} \qquad (17)$$

Usually the value of G is given as a resistance R = 1/G. In such cases we replace G by 1/R in Eqs.(10) to (17).

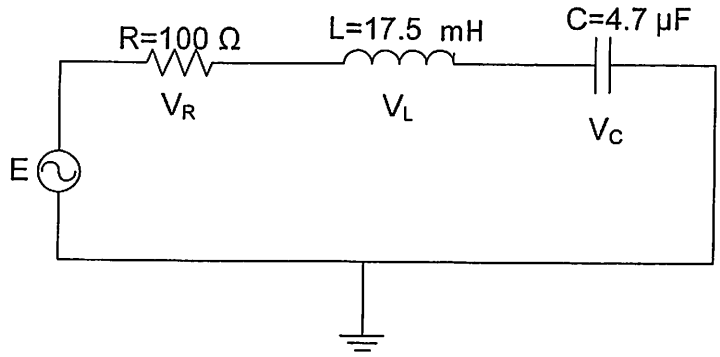


Fig.6:Series resonance

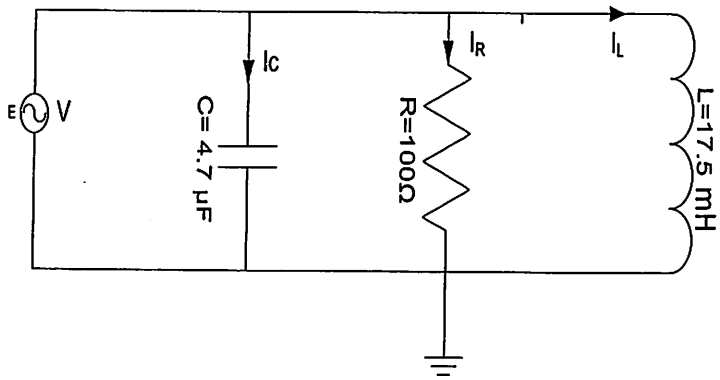


Fig.7:Parsllel resonance

Procedure I: Series RLC Circuit

1. Wire the circuit of Fig. 6 and measure E, I, V_R, V_L, V_C and $Z = E/I$
2. Using impedance diagram prove that $Z = Z_R + Z_L + Z_C$ is equivalent to the measured value of Z in step 1 and calculate θ.
3. Prove that $E = V_R + V_L + V_C$
4. Calculate the Average Power delivered and Power Factor for the circuit of Fig. 6.

Procedure II: Parallel RLC Circuit

1. Wire the circuit of Fig.7 and measure the variables: E, V_R, V_C, V_L, I, I_R, I_C, I_L, Z_L, Z, θ, F_P and P.
   (Hint: Use 10 Ω resistance in series to find θ).
   Note that: Z = V/I.
2. Calculate Z_R, Z_C, Z_L and Z using ohms law.
3. Calculate I using, I = E/Z = EY and compare the result with the measured value.
4. Using CRO, measure the Phase angle between I_C and I_L.
   (Hint: Use 10 Ω resistance, each in series with L and C to find angle)

Procedure III: Series Resonance

1. Wire the circuit shown in Fig.8.

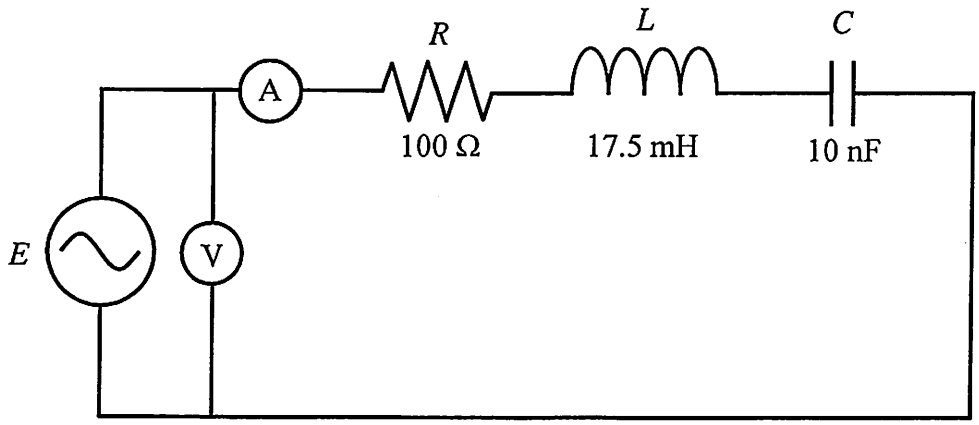


Fig.8: Series resonance.

2. Set the sinusoidal input voltage to $2V_{rms}$ (say) and adjust the frequency to change from 7.5 KHz to 15 Khz in steps of 500 Hz. For each frequency setting fix V to 2 V_{rms}, measure the current I, and record the measured values in Table 1.
3. Calculate $P = I^2 R$ and complete Table 1. Note $P = I^2R$ if I is the rms value and $P = 1/2\ I^2 R$ if I is the peak value.
4. Plot P against frequency. From this plot find the resonant frequency f_{os}, the half-power frequencies f_1 and f_2, the frequency bandwidth B and the quality factor Q_0.
5. Compare the measured f_{os}, f_1, f_2, B and Q_0 with theoretical values calculated from Eqs. 6-a, 6-b, 6-c, 7, and 9, respectively.

Table1:

Frequency Setting $V = 2V_{rms}$	I (mA)	$P = I^2 R$ (mW)
7.500 kHz		
8.000 kHz		
8.500 kHz		
9.000 kHz		
9.500 kHz		
10.000 kHz		
10.500 kHz		
11.000 kHz		
11.500 kHz		
12.000 kHz		
12.500 kHz		
13.000 kHz		
13.500 kHz		
14.000 kHz		
14.500 kHz		
15.000 kHz		

Procedure IV: Parallel Resonance

1. Connect the circuit in Fig. 9.

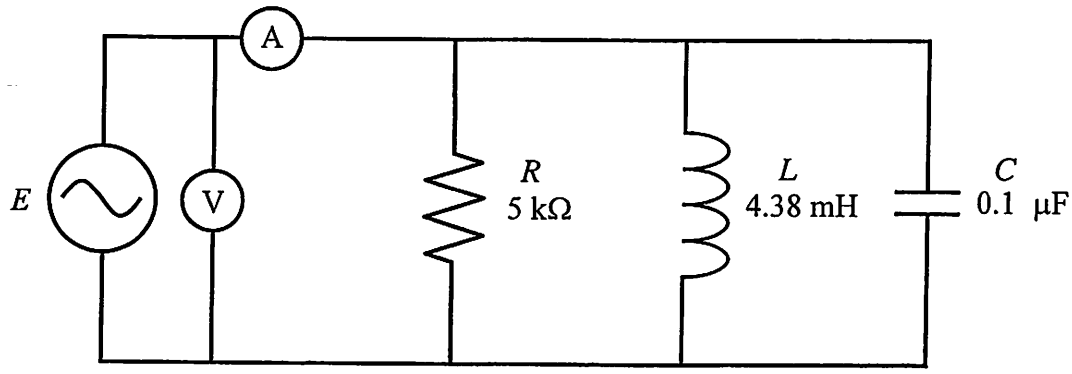


Fig.9:Parallel resonance.

2. Set the sinusoidal voltage to 2 V_{rms} (say) and adjust the frequency to change from 6 KHz to 9 KHz in steps of 200 Hz. For each frequency setting try to fix V at 2 V_{rms} and measure I and record it in Table 2.
3. Calculate Z = V/I and complete Table 2.
4. Plot Z against frequency and deduce from this plot, the resonant frequency f_{op}, the half power frequencies f_1 and f_2, the frequency bandwidth B, and the quality factor Q_0.

Table2:

Frequency Setting $V = 2V_{rms}$	I (mA)	Z = V/I (Ω)
6.000 kHz		
6.200 kHz		
6.400 kHz		
6.600 kHz		
6.800 kHz		
7.000 kHz		
7.200 kHz		
7.400 kHz		
7.600 kHz		
7.800 kHz		
8.000 kHz		
8.200 kHz		
8.400 kHz		
8.600 kHz		
8.800 kHz		
9.000 kHz		